IN-EXTREMIS-MOSTE

Evaluation of Acute Mechanical Revascularization in Large Vessel Occlusion Stroke with Minor Symptoms (NIHSS < 6) in patients last seen well < 24 hours

MinOr Stroke Therapy Evaluation

STUDY CODE: **MOSTE**

Version n°5, 24th June 2024

Sponsored by:
MONTPELLIER CHU

CONTACTS

Sponsor	Montpellier CHU
Coordinating Principal Investigators	**Caroline Arquizan, MD,** Head of Stroke Unit, neurologist. Neurology Department of Montpellier University Hospital Hopital Gui de Chauliac 80, Avenue Augustin Fliche Montpellier, 34000 email: c-arquizan@chu-montpellier.fr Tel : +334 67 33 75 32 Fax: +334 67 33 75 32 **Vincent Costalat, MD, PhD** Head of Neuroendovascular Service - Montpellier University Hospital Professor of Neuroradiology -Montpellier University 80, Avenue Augustin Fliche Montpellier, 34000 email: v-costalat@chu-montpellier.fr Tel : +334 67 33 75 32 Fax: +334 67 33 75 32 **Tudor G. Jovin, MD** Director Cooper Neurological Institute Chair, Department of Neurology Cooper University Healthcare Professor of Neurology and Neurosurgery Cooper Medical School of Rowan University Three Cooper Plaza, Suite 104A Camden, NJ 08103 Email: jovin-tudor@cooperhealth.edu Pittsburgh, PA 1521email: jovitg@upmc.edu Tel: (856) 968-7965 Fax: (856) 968-8222 **Bertrand Lapergue, MD** Head of Stroke Unit, neurologist. Neurology Department of Foch University Hospital Hospital Foch 40, Rue Worth Suresne 92150 email: bertrand.lapergue@gmail.com Tel : +334 67 33 75 32 Fax: +334 67 33 75 32
Study Coordination	ANAGRAM

Statistics & Data Management	Biostatistics and Data Management Department of University of Lille under the responsibility of Professor Alain Duhamel
Safety Contact Data	Montpellier CHU
Study Centers	A current list of sites will be maintained updated in the Study Files

PROTOCOL SYNOPSIS

IN EXTREMIS - MOSTE Trial - Minor Stroke Therapy Evaluation

Title and Code	Evaluation of Acute Mechanical Revascularization in Large Vessel Occlusion Stroke with Minor Symptoms (NIHSS<6) in patients last seen well < 24 hours **M**in**O**r **S**troke **T**herapy **E**valuation**: MOSTE**
Study Design	International, multi-center, prospectively randomized to two parallel (1:1) arms, open treatment, blinded endpoint (PROBE) trial
Blinding	This protocol is designed as an open label treatment assignment
Study Objectives	**Primary Objectives**: To demonstrate the superiority of mechanical thrombectomy (MT) associated with best medical therapy (BMT) compared to BMT alone for increase the functional independence of patients with LVO harboring Minor Stroke Symptoms (NIHSS<6) **Secondary Objectives**: To evaluate the safety and efficacy of MT + BMT vs. BMT alone in patients with LVO harboring Minor Stroke Symptoms (NIHSS<6). In case of neurological deterioration in the BMT arm, secondary mechanical therapy is allowed **Cost-effectiveness study Objective**: Evaluate, in French sites, the cost-effectiveness of MT in patients' subgroup NIHSS 0-5 compared to Standalone Medical Therapy in AIS due to large vessel occlusion.
Sample Size and Duration of Study	A maximum of 824 patients is planned to be recruited in the trial, 412 in the treatment arm and 412 patients in the control arm. Enrollment period: approximately 96 months, until 30/04/2027 Subject participation: 90 days (±14) since randomization Total Study Duration: approximately 99 months
Planned Number of Sites / Countries	Worldwide up 41 sites in participating countries (Europe, USA); each participating site commits to recruit 25 to 40 patients in Europe. US site commits to recruit 10 to 14 patients. There will be up to 10 US sites.

Recruitment of patients (Status)	Enrollment will be monitored through an electronic data capture system
Randomization	Subjects will be randomized 1:1 to mechanical thrombectomy plus best medical treatment or best medical treatment alone. Stratification will occur by: Age < 70 vs ≥ 70 years, occlusion location (ICA vs MCA-M1 or M1-M2 with or without cervical lesion (Tandem)) , IV fibrinolysis, and Symptoms onset or Time Last Known Well 0-4.5, vs > 4.5-23 hours Enrollment in this study is defined as the moment when the randomization process is completed and the subject is assigned to one study arm (time zero)
Analysis Population	Intention to treat analysis (primary and secondary endpoints) 2 interim analyses are planned for efficacy and futility. Separate analyses will be conducted to exclude any subjects treated outside FDA guidelines for usage of thrombolytic therapy and access route for cleared devices.
Study Endpoints	
Efficacy Parameter	Modified Rankin Scale score at 90 days
Primary Efficacy Endpoint	Rate of patients at 90-days with excellent outcome defined as mRS 0-1.
Secondary Efficacy Endpoints	1. Rate of patients with a favorable outcome at 90-days defined as mRS 0-2 2. Rate of patients with perfect functional outcome at 90-days defined as mRS 0 3. Quality of life at 90 days assessed by EuroQol 5D-5L scale 4. Cognitive function at 90-days according to Montreal Cognitive Assessment (MoCA), Trail Making Test A and B, and the STROOP Victoria test 5. Change in NIHSS at 24 (-6/+12) hours post randomization 6. Rate of patients requiring secondary thrombectomy 7. The number of patients with mRS 0-1 at 90 days in the treatment arm versus the number of patients with mRS 0-1 at 90 days in the control arm who underwent secondary MT 8. Infarct volume at 24 (-6/+12) hours from randomization, by cerebral MRI T2/FLAIR or CT 9. Rate of TICI Score 2b/3 assessed by angiography in the treatment arm and in the control arm treated by secondary thrombectomy 10. Revascularization rate at 24 (-6/+12) hours on MRA or CTA
Cost-effectiveness study endpoint	Incremental Cost Effectiveness Ratio (ICER) : difference between the total treatment costs divided by the difference in the outcome (mRS ≤2).
Safety Outcomes	

Primary Safety Endpoint	Incidence of all-cause mortality at 90 days
Secondary Safety Endpoints	Incidence of symptomatic intracerebral hemorrhage according to HEIDELBERG at imaging at 24 (-6/+12) hours post randomization NIHSS ≥ 10 points between admission and D5/D7 Visit (discharge if earlier) Incidence of procedure/device-related adverse events (in patients treated by mechanical treatment, in the MT group or in the BMT group, in case of secondary thrombectomy)
Study Population	Patients who meet all of the inclusion criteria and none of the exclusion criteria (both general and imaging) will be eligible for study participation. All patients considered for participation will have received standard of care stroke treatment prior to eligibility evaluation, including administration of thrombolytic therapy, for patients meeting the institution's guidelines for such treatments. Such treatment will not be delayed due to evaluation for study eligibility.
General Inclusion Criteria	1. Subject is ≥ 18 years old at inclusion (no upper age limit) 2. Clinical signs consistent with acute ischemic stroke with time last known well (TLKW) ≤ 23h[1] at randomization 3. NIHSS 0-5 at the time of randomization with disabling deficit that would prevent a patient from performing basic activities of daily living (bathing, ambulating, toileting, hygiene, and eating) or returning to the same work 4. ASPECT score ≥ 6 on non-contrast CT or DWI-MRI 5. Ischemic Stroke confirmed with cerebral imaging or normal imaging with suspected ischemic stroke 6. Proven anterior circulation intracranial large vessel occlusion on CTA or MRA (ICA, M1, M1-M2) with or without cervical lesion (Tandem) at the time of randomization. The MCA – M1 segment is defined as the first branch of the intracranial ICA which courses horizontally from its branching point off the ICA through the Sylvian fissure up to the first bifurcation distal to the lenticulo-striate arteries, in the Sylvian fissure. The MCA M1-M2 occlusion is considered to represent a distal M1 segment occlusion and is operationally defined as total occlusion of the M2 segment (defined as the portions of the MCA distal to the first bifurcation or trifurcation, but prior to the second bifurcation) and suspicion of any involvement of the M1 segment 7. For Drip and Ship patients: new imaging performed again on inclusion center if first imaging performed > 1 hour before randomization. 8. Patient or patient's representative has received information about the study and has signed and dated the appropriate Informed Consent Form.

[1] *Treatment should be initiated within 24 hours of TLKW (maximum 60min from time of randomization to treatment)*

	9. Anticipated possibility to start the procedure (arterial access) within 60 minutes after randomization 10. Pre stroke mRS ≤ 1 11. For patients for whom thrombolytic therapy, such as IV t-PA, is indicated, regardless of degree of disability, such treatment is initiated as soon as possible and within the accepted clinical guidelines as measured from stroke symptom onset.
General Exclusion Criteria	1. Anticipated impossibility to start the procedure (arterial access) within 60 minutes after randomization 2. Known absence of vascular access 3. Known contrast or endovascular product life-threatening allergy 4. Female who is known to be pregnant or lactating at time of admission 5. Patient who presents with severe or fatal co-morbidities or life expectancy under 6 months that will likely interfere with improvement or follow-up or that will render the procedure unlikely to benefit the patient. 6. Patient unable to be present or available for follow-up 7. Pre-existing neurological or psychiatric disease that would confound the neurological or functional evaluations 8. Evidence of vessel recanalization prior to randomization 9. Seizures at stroke onset if it makes the diagnosis of stroke doubtful and precludes obtaining an accurate baseline NIHSS assessment. 10. Current participation in another investigational drug study 11. Suspicion of aortic dissection based on medical history, clinical evaluation or/and imaging 12. Major patients under guardianship 13. Blood glucose < 50 mg/dL or > 400 mg/dL 14. Cr > 4.0 mg/dL unless the patient is on dialysis 15. Platelet count < 50000/uL 16. *INR > 3.0 or PTT > 3 times upper limit of normal (ULN)*
Imaging Exclusion Criteria	1. Isolated proximal cervical ICA occlusion 2. Evidence of intracranial hemorrhage on CT/MRI 3. Excessive tortuosity of cervical vessels on CTA/MRA that would likely result in unstable access platform 4. High Suspicion of underlying intracranial stenosis on CTA/MRA 5. Suspected cerebral vascular disease (e.g. vasculitis) based on medical history and CTA/MRA 6. Presumed calcified Embolus or Intracranial Stenosis decompensation

	7. Intracranial stent implanted in the same vascular territory that would preclude the safe deployment/removal of the stent retriever device 8. Occlusions in multiple vascular territories (e.g.: bilateral anterior circulation, or anterior circulation/vertebrobasilar system) as confirmed on CTA/MRA 9. Significant mass effect with midline shift as confirmed on CT/MRI
Study withdrawal	1. Death 2. Withdrawal of consent
Endovascular Treatment	MT in the Experimental Arm can be performed with any thrombectomy (CE labeled or FDA cleared for revascularization with code NRY) device usually used at study site and included on the list of devices approved for this study. It is noted that only the Trevo Retriever and Solitaire Revascularization Device have been cleared by the FDA for use in stroke treatment beyond six hours from stroke symptom onset to reduce disability in certain AIS patients. The operator may choose which stent retriever or aspiration catheter to use, depending upon anatomical considerations and personal preference. In cases where a stent retriever is used, the use of a balloon guiding catheter is a technical pre requisite. Additional material, such as aspiration catheter, are encouraged in combination to stent retrievers. No limitation of the number of passes is defined during the procedure however it is <u>strongly recommended not to exceed 3 passes per device or three (3) retrieval attempts in a single vessel.</u> For the subjects randomized to the MT plus BMT arm, the start of treatment is defined as the time of arterial access (for example, groin puncture). Access will be obtained using a transfemoral approach for this study whenever possible; if femoral access is not possible, the details will be documented on the case report form. Appropriate anesthesia, should be administered and achieved as soon as possible, allowing endovascular procedure initiation within 60 minutes after randomization. For the subjects randomized to the Best Medical Treatment arm, **secondary thrombectomy** may **be performed within 24 hours of randomization only if neurological deterioration occurs**. This is defined as worsening in the NIHSS of ≥ 4 points compared to baseline or reaching NIHSS of ≥6, or if there is a clinical deterioration worth intervention according to the attending physician. To qualify for such a “secondary” thrombectomy, the patient will

	undergo a rescreening process to ensure patient safety. A non-contrast CT scan will be performed to check for hemorrhage or other non-ischemic causes of stroke progression that would disqualify the patient from such treatment. An ASPECT score > 6 on the rescreening CT is needed to meet eligibility requirements. Patients who deteriorate beyond 24 hours of randomization should be treated according to local standards of care. The change in NIHSS value and the chosen treatment should be recorded in the eCRF by certified trained research staff. Prior to the start of the endovascular procedure, the modified TICI score should be assessed and recorded The late venous phase should be included in all angiogram acquisitions. Embolization to New Territory (ENT): any new infarct on neuro-imaging at 24 (-6/+12) hours compared to baseline images, wherever catheterization was performed Concomitant Medications in patients having EVT: • Use of IA lytics, is not allowed (protocol deviation) during the procedure and until after follow up imaging is completed. • Systemic anticoagulation with heparin may be used during the procedure, but should not exceed a total of 3,000 units of heparin bolus followed by a 500 units/hour drip for the duration of the procedure. • Prudent use of anti-vasospasm agents is permitted during the procedure. • For patients having carotid stenting, antiplatelet agents are allowed at the discretion of the investigator.

Best Medical Treatment	The administration of medications is at the treating physician's discretion. Subjects previously treated with *anticoagulant and/or* antiplatelet agents, may continue this regimen if, in the investigator's opinion, the benefits of continued therapy outweigh the risks of potential neurological deterioration related to hemorrhage. All subjects enrolled into this study should be medically managed according to the current European Stroke Organization or AHA/ASA guidelines. Separate data analyses will be performed for any cases in which thrombolytic therapy is used beyond the FDA-approved three hour time window from symptom onset, with such cases omitted from the regulatory data analysis.
Withdrawal and Replacement of Subjects	Participant may withdraw from the study at any time, with or without reason and without prejudice to further treatment or medical care. Subjects may be also withdrawn from the study at the discretion of the PI or Sponsor for safety or administrative reasons
Follow-Up Schedule per patient	All follow up time points are relative to time of randomization (time zero) Visit 1: Baseline, containing all required baseline data Visit 2: 24 (-6/+12) hours, NIHSS and MRI/MRA or CT/CTA. Follow-up hemorrhage and infarct volume will be measured by MRI T2/Flair or CT. Short IQCode. Visit 3: 5-7th day (if subject remains in hospital) or discharge, whichever is earlier: NIHSS evaluation Visit 4: 90th day (±14), blinded mRS, EuroQol 5D-5L,QALY, MoCA, Trail Making tests A and B evaluations, and STROOP Victoria test (French hospitals)

PROTOCOL SIGNATURE PAGE

The signatures on this page indicate review and approval of the final version of the protocol.

STUDY TITLE: Minor Stroke Therapy Evaluation

STUDY CODE: MOSTE
PROTOCOL VERSION: V5 – 24th June 2024

By signing this document we confirm that the clinical study will be conducted in accordance with the protocol and all applicable laws and regulations including, but not limited to, the International Conference on Harmonization Guideline for Good Clinical Practice (GCP) and the ethical principles that have their origins in the Declaration of Helsinki.

I agree to personally conduct or supervise the research, and ensure all participating investigators and research staff are appropriately informed and trained prior to participating in any study related activities.

Renan TARGHETTA CHU Montpellier's Representative Name	Signature	Date of signature
Dr. Caroline Arquizan Coordinating Principal Investigator	Signature	Date of signature
Pr. Vincent Costalat Coordinating Principal Investigator	Signature	Date of signature
Dr. Bertrand Lapergue Coordinating Principal Investigator	Signature	Date of signature
Dr. Tudor Jovin Coordinating Principal Investigator	Signature	Date of signature

LOCAL PI SIGNATURE PAGE

The signatures on this page indicate review and approval of the final version of the protocol.

STUDY TITLE: Evaluation of Acute Mechanical Revascularization in Large Vessel Occlusion Stroke with Minor Symptoms (NIHSS < 6) in patients last seen well < 24 hours

STUDY CODE: MOSTE
PROTOCOL VERSION: V5 – 24th June 2024

By signing this document I confirm that the clinical study will be conducted in accordance with the protocol and all applicable laws and regulations including, but not limited to, the International Conference on Harmonization Guideline for Good Clinical Practice (GCP) and the ethical principles that have their origins in the Declaration of Helsinki.

I agree to personally conduct or supervise the research, and ensure all participating investigators and research staff are appropriately informed and trained prior to participating in any study related activities.

Principal Investigator Name	Signature	Date of signature

INDEX

CONTACT DETAILS	**2**
PROTOCOL SYNOPSIS	**4**
PROTOCOL SIGNATURE PAGE	**11**
LOCAL PI SIGNATURE PAGE	**12**
INDEX	**13**
Introduction and Rationale	**15**
1. Study Design	**19**
2. Blinding	**19**
3. Study Objectives	**19**
3.1 Primary Objectives	**19**
3.2 Secondary Objectives	**20**
3.3 Medico-economic study Objective	**20**
4. Sample Size and Duration of Study	**20**
5. Planned Number of Sites / Countries	**20**
6. Recruitment of patients (Status)	**20**
7. Randomization	**20**
8. Analysis Population	**21**
9. Study Endpoints	**21**
9.1 Efficacy Parameter:	21
9.2 Primary Efficacy Endpoint	22
9.3 Secondary Efficacy Endpoints	22
9.4 Medico-economic study Endpoints	22
10. Safety Outcomes	**20**
10.1 Primary safety Endpoint:	20
10.2 Secondary Safety Endpoints	20
11. Study Population	**23**
11.1 General Inclusion Criteria	23
11.2 General Exclusion Criteria	24
11.3 Imaging Exclusion Criteria	25
11.4 Study Withdrawal	25
12. Endovascular Treatment	**26**
13. Best Medical Treatment	**32**
14. Both Arms	**32**
15. Withdrawal and Replacement of Subjects	**33**
16. Follow-up Schedule per patient	**33**
Figure 1. MOSTE Trial Flow Chart	33
Table 1. MOSTE Trial Schedule of VISITS / EVALUATIONS	34
16.1 Informed Consent	35
16.2 Before Randomization Procedure	35
16.3 24 (-6/+12) Hours post Randomization	36

16.4 Day 5-7 / Discharge	37
16.5 Day 90 (± 14)	37
17. Statistical Methods	**38**
17.1 Sample Size Calculation	38
17.2 Loss of follow-up and Missing Data	39
17.3 Statistical Analysis	39
18. Data Collection and Management	**42**
19. Monitoring Procedures	**43**
19.1 Auditing or Inspection	44
19.2 Site and Investigator selection	44
19.3 Investigator Responsibilities	44
19.4 Training of Investigator and Site Personnel	45
20. Adverse Events	**45**
20.1 Adverse Event Definitions and Classification	45
20.2 Serious and not serious expected adverse reactions	46
20.3 Events of interest	47
20.4 Investigator responsibilities	48
20.5 Sponsor Responsibilities	**51**
21. Risk Benefit Analysis	**53**
21.1 Risk Minimization	54
22. Study Committees, Core Labs and Partner	**55**
22.1 Steering Committee	55
22.2 Safety Monitoring Committees	56
22.3 Clinical Events Committee (CEC)	57
22.4 Imaging lab cores	57
22.5 ARN GDC	59
23. Ethical Considerations	**60**
23.1 Compliance with Good Clinical Practices (GCP) and Adherence to Protocol	60
23.2 Institutional Review Board/ Ethics Committee	60
23.3 Insurance	**61**
24. Study Administration	**62**
24.1 Pre-Study Documentation Requirements	62
24.2 Record Retention	62
24.3 Criteria for Terminating Study	62
24.4 Criteria for Suspending/Terminating a Study Site	62
25. Recommendations	**63**
26. References	**66**
Appendix A. Abbreviations	**69**
Appendix B Definitions	**72**

Appendix C. Angiographic Core Lab Scales	**75**
Appendix D. Medico-economic study	**77**
Appendix E. Adverse Events Management in Spain	**83**

Introduction and Rationale

Stroke represents the fourth leading cause of death in industrialized nations, after heart disease, cancer, and chronic lower respiratory disease. The overall burden of stroke will rise dramatically in the next 20 years due to an aging population. The projections in the Burden of Stroke report indicate that between 2015 and 2035, overall there will be a 34% increase in total number of stroke events in the European Union from 613,148 in 2015 to 819,771 in 2035[1].

Approximately one-quarter of the patients suffering a stroke die within one year after the initial event and stroke is a leading cause of serious long-term disability. Direct medical costs related to stroke in the United States is an estimated $28.3 billion per year[3]. In the EU the total cost of stroke in 2015 was calculated as €45 billion[1]. We can then conclude that stroke brings a dramatic financial and personal burden to society.

Proximal intracranial arterial occlusions cause the most disabling types of ischemic strokes and are predictive of poor neurological outcomes[2]. The “Time is Brain” assessment has been confirmed in many recent trials (MR CLEAN[27], SWIFT PRIME[28], REVASCAT[29], HERMES[30]) within the first 6 hours. Recently the time window has been enlarged to 24h00 after the DAWN trial results[31], enforcing nevertheless the strong relation between patient disability and time loss in this extended time window.

In population-based studies, patients presenting with minor or mild stroke symptoms represent about two-thirds of stroke patients, and almost one-third of these patients are unable to ambulate independently at the time of discharge[4,5].

Although mechanical thrombectomy (MT) has become the standard of care for acute ischemic stroke with proximal large vessel occlusion (LVO) in the anterior circulation, the management of patients harboring proximal occlusion but presenting minor-to-mild stroke symptoms, has not yet been determined by these recent randomized clinical trials[6,7]. Indeed, the majority of patients presented with major clinical impairment, with a median NIHSS of 17[8]. Thus, American

Heart Association (AHA) gives level 1a evidence for MT performed only for patients with baseline NIHSS score ≥6[9].

However, patients with proximal occlusions may present with a low NIHSS[10], a proximal intra-arterial occlusion being present in up to 28% when considering patients with an NIHSS ≤ 4[11]. In an observational study, patients with minor or mild stroke symptoms and LVO have a high risk of both clinical worsening and bad outcome[12].

The STAIR (Stroke Treatment Academic Industry Roundtable) meeting aims to advance acute stroke therapy development through collaboration between academia, industry, and regulatory institutions[13]. In pursuit of this goal and building on the available level I evidence of benefit from endovascular therapy (EVT) in large vessel occlusion stroke, STAIR IX consensus recommendations were developed that outline priorities for future research in EVT. Three key directions for advancing the field were identified:

(1) Development of systems of care for EVT in large vessel occlusion stroke

(2) Development of therapeutic approaches adjunctive to EVT

(3) Exploring clinical benefit of EVT in patient population insufficiently studied in recent trials and explore whether patient eligibility could be expanded

Recent AHA/ASA guidelines have also highlighted the need to gain more evidence to determine whether there is an overall net benefit from endovascular therapy (EVT) in patients with LVO and minor stroke[9].

The effectiveness and safety of MT, in the subgroup of minor stroke with LVO in the anterior circulation, is still open to debate. Data about MT in this subgroup of patients are sparse, and their optimal management has not yet been defined[14].

Previous Studies

A retrospective study conducted in France has analyzed the data extracted from 3 prospective clinical registries of all consecutive patients with acute cerebral infarct and minor-to-mild symptoms treated by MT, with or without intravenous thrombolysis, between January 2012 and March 2016[15]. The purpose of this study was to evaluate the impact of reperfusion on functional outcome after MT. Regarding patient selection, it corresponds to those used in several studies, both for minor (NIHSS ≤ 3)[16,17] and mild strokes (NIHSS <8)[18].

The study shows that the reperfusion status achieved with MT strongly impacts the functional outcome among minor to-mild stroke patients with LVO. The excellent outcome (mRS 0-1) rate increased with reperfusion grades, with 34.6% of patients with no successful reperfusion having excellent outcome in comparison to 61.7% in patients with TICI 2B reperfusion and 78.5% in patients with TICI 3 reperfusion (p for trend <0.001). There are already known the limitations inherent to a retrospective design, added to a relatively small sample size extracted from 3 different centers but, the main limitation is the absence of a control group undergoing optimal medical management alone.

One previous study with a relatively small sample size (n = 33) has already shown that clinical outcome is predominately favorable for minor stroke patients having undergone MT[19]. Excellent and favorable outcomes were achieved in 42.4 and 63.6%[19].

Another recent small case–control study (n=10 in the MT group and 22 in the medical group) suggested that MT may lead to a shift towards a lower NIHSS at discharge than due to medical management alone (rate of excellent outcome, MT, 70% versus BMT, 55%)[20]. An analysis from STOPStroke and GESTOR cohorts (n=30 in the MT group and 88 in the medical group) similarly concludes that in patients presenting with minimal stroke symptoms (NIHSS score ≤5) and large vessel occlusion strokes, MT appears to be associated with a favorable shift of NIHSS at discharge, as well as higher rates of independence at discharge and long-term follow-up[21].

Another retrospective study included 378 patients with minor strokes in the anterior circulation; 54 (14.2%) of these had proven large-vessel occlusions. Eight of 54 (14.8%) were immediately treated with mechanical thrombectomy, 6/54 (11.1%) after early neurologic deterioration, and the rest were treated with standard thrombolysis only. Rates of successful recanalization were similar between the 2 mechanical thrombectomy groups (75% versus 100%). Rates of excellent outcome (modified Rankin Scale 0–1) were higher in patients with immediate thrombectomy (75%) compared with patients with delayed thrombectomy (33.3%) and thrombolysis only (55%). No symptomatic intracranial hemorrhage occurred in either group[22].

A multicenter cohort study of minor and mild AIS patients harboring LVO in the anterior circulation, conducted in 4 comprehensive stroke centers with two therapeutic approaches (urgent MT associated with BMT versus BMT first and MT if worsening occurs) included 301 patients (170 with urgent MT associated with BMT and 131 with BMT alone as first line treatment). The primary end-point was the rate of excellent outcome defined as the achievement of a mRS score of 0–1 at 3 months[23]. In an intention-to-treat analysis, patients treated with urgent MT were younger, more often received intravenous thrombolysis, and had shorter time to imaging. Twenty-four patients (18.0%) in the BMT group had rescue MT due to neurologic worsening. Overall, excellent outcome was achieved in 64.5% of patients, with no difference between the two groups. Stratified analysis according to key subgroups did not find heterogeneity in the treatment effect size. Study conclusion was that minor-to-mild stroke patients with LVO achieved excellent and favorable functional outcomes at 3 months in similar proportions between urgent MT versus delayed MT associated with BMT.

SUMMARY of published studies:

Reference of publication	**Dargazanli C et al, Stroke 2017**	**Dargazanli C et al, Stroke 2017**	**DC HAUSSEN JNIS 2017**	**DC HAUSSEN JNIS 2017**	**MP MESSER AJNR 2017**	**MP MESSER AJNR 2017**	**MP MESSER AJNR 2017**	**HERMES**	**HERMES**	**HERMES**
Sample size: number of patients	92	113	88	30	8	6	40			
Treatment Approach	BMT*	BMT*+EVT**	BMT*	BMT*+EVT**	BMT*+EVT**	BMT*+ delayed EVT**	BMT*	BMT*+EVT**	BMT*	BMT*+ rescue EVT**
Baseline NIHSS	0-5	0-5	0-5	0-5	0-5	0-5	0-5	0-10	0-10	0-10
mRS 0-1 (%)	70.7	67.3	60	68	75	33	55	75	39	33

*BMT (Best Medical Treatment), **EVT (Endovascular Therapy)

Overall conclusion leads to the prediction that successful reperfusion allows better functional outcome at 3 months in patients presenting with minor-to mild stroke symptoms harboring proximal arterial occlusion in the anterior circulation. According to these observational studies, there is thus an urgent need for randomized controlled trials to define the effectiveness and safety of MT in patients with LVO and minor deficit.

MOSTE study has been designed with subject safety in mind, in order to address the concerns around potential unknown harms to enrolled subjects. If this study is positive, more patients in the future could receive urgent endovascular treatment in addition to the best medical treatment.

1. Study Design

The MOSTE protocol is an International, multi-center, prospectively randomized to two parallel (1:1) arms, open to treatment with blinded endpoint trial, designed to demonstrate that mechanical thrombectomy with best medical treatment is superior to medical treatment alone, in improving clinical outcomes at 90 days, in patient presenting an acute large vessel occlusion stroke with a minor deficit, defined as NIHSS below 6 and < 24 hours from onset.

2. Blinding

This protocol is designed as an open label treatment assignment.
Review of all images and calculations of extension/volume will be conducted by the CT/MR core lab, on an ongoing basis to provide consistent, independent evaluation of images for confirmation inclusion criteria, and independent assessment of angiograms. Each site must designate one or more individual(s) to perform the blinded mRS assessments at Day 90 (±14). This individual will be identified on the Authorization Signature Log and must not perform data entry or other tasks that would reveal the study arm assignment of subjects. Moreover, the blinded evaluator(s) will be instructed to follow a scripted interview to minimize the chance of subjects disclosing their treatment group to the evaluator, and will also be required to self-certify that they remained blinded throughout the interview with the subject.

3. Study Objectives

3.1. Primary Objectives

To demonstrate the superiority of mechanical thrombectomy associated with best medical therapy compared to best medical therapy alone for increase the functional independence of patients with LVO harboring Minor Stroke Symptoms (NIHSS<6).

3.2. Secondary Objectives

To compare the efficacy (assessed by other clinical endpoints) and the safety between the two therapeutic approaches (MT associated with best medical therapy vs best medical therapy alone) in LVO patients harboring Minor Stroke Symptoms (NIHSS<6).

To describe, in all patients treated by MT (first line or secondary thrombectomy in case of deterioration in the BMT group), the angiographic efficacy outcomes and procedure-related–adverse events.

3.3. Medico-economic study Objective

A medico-economic study will be performed for data from French sites. The details of this analysis are available in a separate document

4. Sample Size and Duration of Study

- A maximum of 824 patients is planned to be recruited in the trial, 412 in the treatment arm and 412 patients in the control arm.
- Enrollment period is approximately 96 months, until 30/04/2027. Subject participation: 90 days (±14) since randomization.
- Subject participation: 90 days (±14) since randomization (time zero)
- Total Study Duration: approximately 99 months

5. Planned Number of Sites / Countries

Worldwide up 41 sites in participating countries (Europe, USA); each participating site commits to recruit 25 to 40 patients in Europe. Each participating US site commits to recruit 10 to 14 patients. There will be up to 10 US sites.

6. Recruitment of patients (Status)

Enrollment will be monitored through an electronic data capture system which will be set to automatically notify the CRA or Project Manager of all subject enrollments at real time when entered within the system.

7. Randomization

If the subject's eligibility status is confirmed, patients will be randomly allocated in a one-to-one ratio to receive best medical therapy combined with mechanical treatment (experimental arm) or best medical therapy alone (control arm). To assure a centralized real time randomization procedure, a web-based randomization will be performed using the electronic case-report form (eCRF) system. The patient randomization numbers will be allocated sequentially in the order in which the patients are randomized. A dynamic randomization procedure by minimization will be done for achieving a balance of the following pre-specified factors: age (70≤ vs >70), occlusion site (ICA vs MCA-M1 or M1-M2 with or without cervical lesion (Tandem)), IV fibrinolysis and time (symptoms onset to randomization) 0-4.5 hours, vs 4.5-23 hours. The center will be also considered in the minimization method. Patients are enrolled and randomized by vascular neurologists and neurointerventionalists.

The automated randomization system, will inform via automatic pre-programmed notifications within the system and to sponsor's designees, when a patient is randomized.

Enrollment in this study is defined as the moment when the randomization process is completed and the subject is assigned to a study arm (time zero).

8. Analysis Population

- Intention to treat analysis (primary and secondary endpoints)
- 2 interim analyses are planned for efficacy and futility
- Separate analyses will be conducted to exclude from the regulatory data analysis any subjects treated outside FDA guidelines for usage of thrombolytic therapy and of cleared devices.

9. Study Endpoints

9.1. Efficacy Parameter:

Modified Rankin Scale score at 90 days

9.2. Primary Efficacy Endpoint:

The primary endpoint is the rate of patients with excellent functional outcome at 90 days defined by a modified Rankin Scale (mRS) 0 or 1.

9.3. Secondary Efficacy Endpoints

- Rate of patients with a favorable outcome at 90-days defined as mRS 0-2
- Rate of patients with perfect functional outcome at 90-days defined as mRS 0
- Quality of life at 90 days assessed by EuroQol 5D-5L scale
- Cognitive function at 90-days according to Montreal Cognitive Assessment (MoCA), Trail Making Test A and B, and the STROOP Victoria test
- Change in NIHSS at 24 (-6/+12) hours post randomization
- Rate of patients requiring secondary thrombectomy
- The number of patients with mRS 0-1 at 90 days in the treatment arm versus the number of patients with mRS 0-1 at 90 days in the control arm who underwent secondary thrombectomy
- Infarct volume growth at 24 (-6/+12) hours from randomization, by cerebral MRI T2/FLAIR or CT
- Rate of TICI Score 2b/3 assessed by angiography in the treatment arm and in the control arm treated by secondary thrombectomy
- Revascularization rate at 24 (-6/+12) hours on MRA or CTA

9.4. Medico-economic study Endpoints

These endpoints are listed in a separate document.

10. Safety Outcomes

10.1. Primary safety Endpoint:

Incidence of all cause mortality at 90 days

10.2. Secondary Safety Endpoints

- Incidence of SICH, by HEIDELBERG definition, at 24 (-6/+12) hours post randomization

- NIHSS ≥ 10 points between admission and D5/D7 Visit (discharge if earlier)
- Incidence of procedure/device-related adverse events (in patients treated by mechanical treatment, in the MT group or in the BMT group, in case of secondary therapy) :
  - Vascular perforation
  - Arterial dissection
  - Embolization to a new territory
  - Access site complication requiring surgical repair or blood transfusion
  - Intra-procedural mortality
  - Device failure (in vivo breakage)
  - Any other complications adjudicated by the CEC to be related to the procedure

11. Study Population

Patients who meet all of the inclusion criteria and none of the exclusion criteria will be eligible for study participation. All patients considered for participation will have received standard of care stroke treatment prior to eligibility evaluation, including administration of thrombolytic therapy, for patients meeting the institution's guidelines for such treatments. Such treatment will not be delayed due to evaluation for study eligibility.

11.1. General Inclusion Criteria

- Subject is ≥ 18 years old at inclusion (no upper age limit)
- Clinical signs consistent with acute ischemic stroke with time last known well (TLKW) ≤ 23h at randomization (With the goal of remaining within 24 hours from TLKW to treatment)
- NIHSS 0-5 at the time of randomization with potentially disabling deficit that would prevent a patient from performing basic activities of daily living (bathing, ambulating, toileting, hygiene, and eating) or returning to the same work

- ASPECT score ≥ 6 on non-contrast CT or DWI-MRI
- Ischemic Stroke confirmed with cerebral imaging or normal imaging with suspected ischemic stroke
- Proven anterior circulation intracranial large vessel occlusion on CTA or MRA (ICA, M1, M1-M2 with or without cervical lesion (Tandem)) at the time of randomization.The MCA – M1 segment is defined as the first branch of the intracranial ICA which courses horizontally from its branching point off the ICA through the Sylvian fissure up to the first bifurcation distal to the lenticulo-striate arteries, in the Sylvian fissure. The MCA M1-M2 occlusion is considered to represent a distal M1 segment occlusion and is operationally defined as total occlusion of the M2 segment (defined as the portions of the MCA distal to the first bifurcation or trifurcation, but prior to the second bifurcation) and suspicion of any involvement of the M1 segment.
- For Drip and Ship patients: New imaging performed again on inclusion center if first imaging performed > 1 hour before randomization.
- Patient or patient's representative has received information about the study and has signed and dated the appropriate Informed Consent Form
- Anticipated possibility to start the procedure (arterial access) within 60 minutes after randomization
- Pre stroke mRS ≤ 1
- For patients for whom thrombolytic therapy, such as IV t-PA, is indicated, regardless of degree of disability, such treatment is initiated as soon as possible and within the accepted clinical guidelines as measured from stroke symptom onset.

11.2. General Exclusion Criteria

- Anticipated impossibility to start the procedure (arterial access) within 60 minutes after randomization
- Known absence of vascular access
- Known contrast or endovascular product life-threatening allergy
- Female who is known to be pregnant or lactating at time of admission

- Patient who presents severe or fatal co-morbidities or Life expectancy under 6 months that will likely interfere with improvement or follow-up or that will render the procedure unlikely to benefit the patient.
- Patient unable to present or be available for follow-up
- Pre-existing neurological or psychiatric disease that would confound the neurological or functional evaluations
- Evidence of vessel recanalization prior to randomization
- Seizures at stroke onset if it makes the diagnosis of stroke doubtful and precludes obtaining an accurate baseline NIHSS assessment.
- Current participation in another investigational drug study
- Suspicion of aortic dissection based on medical history, clinical evaluation or/and imaging
- Major patients under guardianship
- Blood glucose < 50 mg/dL or > 400 mg/dL
- Cr > 4.0, unless the patient is on dialysis
- Platelet count < 50000/uL
- INR > 3.0 or PTT > 3 times upper limit of normal (ULN)

11.3. Imaging Exclusion Criteria

- Evidence of intracranial hemorrhage on CT/MRI
- Excessive tortuosity of cervical vessels on CTA/MRA that would likely result in unstable access platform
- High suspicion of underlying intracranial stenosis on CTA/MRA
- Suspected cerebral vascular disease (e.g. vasculitis) based on medical history and CTA/MRA
- Presumed calcified Embolus or Intracranial Stenosis decompensation
- Intracranial stent implanted in the same vascular territory that would preclude the safe deployment/removal of the stent retriever device

- Occlusions in multiple vascular territories (e.g.: bilateral anterior circulation, or anterior circulation/vertebrobasilar system) on CTA/MRA
- Significant mass effect with midline shift as confirmed on CT/MRI

11.4. Study withdrawal

- Death
- Withdrawal of consent

12. Endovascular Treatment

MT in the Experimental Arm can be performed with any thrombectomy (CE labeled or FDA cleared for revascularization with code NRY) device usually used at study site and included on the list of devices approved for this study. It is noted that only the Trevo Retriever and Solitaire Revascularization Device have been cleared by the FDA for use in stroke treatment beyond six hours from stroke symptom onset to reduce disability in certain AIS patients).. During the study procedure, the operator may choose which stent retriever or aspiration catheter to use, depending upon anatomical considerations and personal preference. In cases where a stent retriever is used the use of a balloon-guiding catheter is a technical pre requisite. **Additional material such as aspiration catheter, are strongly encouraged in combination to stent retrievers**. No limitation of the number of passes is defined during the procedure and it is left to the judgment of the practitioner but, **it is <u>strongly recommended not to exceed 3 passes per device or three (3) retrieval attempts in a single vessel</u>**.

For the subjects randomized to the Mechanical Thrombectomy plus medical management arm, start of treatment is defined as the date and time of arterial access (for example, groin puncture).

Access will be obtained using a transfemoral approach for this study whenever possible; if femoral access is not possible, the details will be documented on the case report form.

Appropriate anesthesia, should be administered following the standard practice at the investigational site, and should be achieved as soon as possible, allowing endovascular procedure initiation within **60 minutes after randomization**. The occlusion location(s) will be recorded by the site on the corresponding eCRF.

For the subjects randomized to the Best Medical Treatment arm, **secondary thrombectomy** may be performed within 24 hours of randomization **only** if neurological deterioration occurs. This is defined as worsening in the NIHSS of ≥ 4 points compared to baseline or an NIHSS of ≥6, or if there is a clinical deterioration worth intervention according to the attending physician. To qualify for such a “secondary” thrombectomy, the patient will undergo a rescreening process to ensure patient safety. A non-contrast CT scan will be performed to check for hemorrhage or other non-ischemic causes of stroke progression that would disqualify the patient from such treatment. An ASPECT score > 6 on the rescreening CT is needed to meet eligibility requirements.

Patients who deteriorate beyond 24 hours of randomization should be treated according to local standards of care. **The change in NIHSS value and the chosen treatment should be recorded in the eCRF by certified trained research staff.**

Angiographic evaluations: need to be done and recorded at these timelines, before device use, after device use, and post procedure to determine vessel patency, the presence of embolization to new territory (ENT) or distal emboli (DE), and contrast extravasation/possible intracranial hemorrhage.

Prior to the start of the endovascular procedure, the **modified TICI score** within the vascular territory being treated should be assessed. Angiographic films of the LVO to be treated must allow clear visualization of the target artery. The same orientation should be used before and after the Mechanical Thrombectomy in order to allow a valid analysis of the reperfusion status of the vessel(s).

The **late venous phase** should be included in all angiogram acquisitions.

Embolization to New Territory (ENT) is: defined as any new infarct on neuro-imaging at 24 (-6/+12) hours compared to baseline images, wherever catheterization was performed. Any new neurological deterioration occurring post EV procedure and not secondary to the initially affected area (with or without CT/MRI lesion equivalent) will also be adjudicated as ENT.

In the event of a **procedural complication** or adverse event, detailed angiographic images should be obtained and sent to Core Lab or CEC upon request.

For ALL subjects included and randomized in the MOSTE trial, **Adverse Events** and corresponding classification and action/s taken, as well as **concomitant treatments**, must be reported and recorded in the corresponding panels of eCRF in an ongoing basis, during all portions of the study period (from time of randomization).

Sites should submit all angiographic data to the Core Lab, rather than pre-selecting a subset of images. If angiographic images are missing from the sequence of acquisitions, the core lab will request the site to resend the angiography dataset.

Sensitivity analyses will be performed excluding subjects who do not have a protocol specified lesion and are treated off protocol with any other device than approved stent retrievers.

Of note, for randomized subjects who are randomized to the experimental arm, but who are not treated with endovascular therapy due to spontaneous recanalization or other factors (inability to access the lesion, etc.) the subject is considered enrolled and the site must still follow the subject through 90 days and collect all relevant data.

Concomitant Medications in patients having EVT

- Use of IA lytics, is not allowed (protocol deviation) during the procedure
- Systemic anticoagulation with heparin may be used during the procedure, but should not exceed a total of 3,000 units of heparin bolus followed by a 500 units/hour drip for the duration of the procedure.
- Prudent use of anti-vasospasm agents is permitted during the procedure
- For patients having carotid stenting, antiplatelet agents are allowed at the discretion of the investigator.

Unexpected Angiographic Findings

One of the main inclusion criteria for the study is the presence of an Intracranial ICA and/or MCA-M1 or M1-M2 segment occlusion on the pre-randomization CTA/MRA.

Subject with isolated proximal cervical ICA occlusions, isolated M2 occlusions and subjects with occlusions in multiple vascular territories (e.g., bilateral anterior circulation, or anterior/posterior circulation) **on the pre-randomization neuro-imaging** are excluded from the study.

Due to the high accuracy of CTA/MRA in detecting proximal LVO, a close correlation with the findings on conventional angiography is expected; however, the following unexpected situations may be found:

NO intracranial arterial proximal occlusion in any treatable vessel on the diagnostic angiography, no stent retriever device will be used and the procedure will be ended.

After review of the neuro-imaging by the Core Labs, the data from these cases will be analyzed according to one of the following rules:

1. If the occlusion was misdiagnosed by the investigational site on the initial neuro-imaging evaluation (determined by Core Lab evaluation), this will be considered a major protocol violation and the subject analyzed in the ITT population analysis, using their "actual" mRS score as the primary outcome measure.

2. If the occlusion was present on the initial neuro-imaging evaluation (determined by Core Lab evaluation) but is not visualized on the baseline diagnostic angiogram by the Angiographic Core Lab, these subjects will be categorized as having achieved spontaneous recanalization and will be analyzed in both the "intent-to-treat analysis" and the "per protocol" analysis, utilizing their "actual" mRS score as the primary outcome measure.

Additional sensitivity analyses will be performed excluding subjects who do not receive the assigned therapy.

If thrombus is identified in one or more proximal non treatable arteries per protocol and in none of the per-protocol treatable arteries on the initial diagnostic angiography (e.g. cervical ICA, anterior cerebral artery (ACA), posterior cerebral artery (PCA), vertebral artery (VA) or basilar artery (BA)), these occlusions may be treated as per local standards.

After review of the multimodal CT/MRI and Angiograms by the Core Labs, the data from these cases will be analyzed according to one of the following rules:

1. if the occlusion location (enrollment criteria) was misdiagnosed by the enrolling center on the initial CTA/MRA evaluation (as per CT/MR Core Lab determination), this will be considered a major protocol violation and these subjects will be analyzed in the “intent-to-treat” analysis, **but not the “per protocol”** analysis, utilizing their “actual” mRS score as the primary outcome measure.

2. If there is a new occlusion present in a non-treatable vessel per protocol that was not present on the initial CTA/MRA evaluation (as per CT/MR Core Lab determination), this will be considered a **procedure-related serious adverse event** (e.g. Embolization to a new territory) and these subjects will be analyzed in both the “intent-to-treat” analysis and the “per protocol” analysis, utilizing their “actual” mRS score as the primary outcome measure.

Sensitivity analyses will be performed excluding subjects who do not have a protocol specified lesion and are treated off protocol with any device.

If thrombus is identified in one or more distal non treatable arteries, per protocol (e.g. the ipsilateral M2 or M3 MCA segment), and none of the per protocol treatable arteries on the initial diagnostic angiography, and the occlusion in the opinion of the physician caring for the subject could potentially lead to major disability, it may be treated as per the local management guidelines.

After review of the multimodal CT/MRI and Angiograms by the Core Labs, the data from these cases will be analyzed according to one of the following rules:

1. If a treatable occlusion was present on the initial CTA/MRA evaluation (as per CT/MR Core Lab determination), but is not visualized on the baseline diagnostic Angiogram by the Angiographic Core Lab, these subjects will be categorized as having achieved spontaneous recanalization with distal clot migration and will be analyzed in both the

"intent-to-treat analysis" and the "per protocol" analysis, utilizing their "actual" mRS score as the primary outcome measure.

Sensitivity analyses will be performed excluding subjects who do not have a protocol specified lesion and are treated off protocol with any other device than approved stent retrievers.

Of note, for randomized subjects who are randomized to the treatment arm, **but who are not treated with endovascular therapy due to spontaneous recanalization or other factors** (inability to access the lesion, etc.), the subject is considered enrolled and the site must still follow the subject through 90 days and collect all relevant data

End of the EV Procedure

The EV procedure should be ended if any of the following circumstances occurs:

- Neurological deterioration or alteration in function leading to the suspicion of an intracranial hemorrhage will require an emergent head CT or MRI scan. At the discretion of the investigator, this evaluation may also include angiography or other diagnostic tests to determine the etiology of the clinical alteration. Management of an intracranial hemorrhage will be performed according to each institution's usual practice.
- TICI grade 2b or 3 flow (reperfusion of > 1/2 MCA territory) is established (according to the modified TICI score)
- The occlusion is refractory to at least three (3) retrieval attempts in a single vessel (3 attempts each) at the discretion of the investigator
- Two different devices can be used among the stent retrievers or aspiration catheter family

Neurological deterioration or alteration in function leading to the suspicion of an intracranial hemorrhage will require an emergent head CT or MRI scan. At the discretion of the investigator, this evaluation may also include angiography or other diagnostic tests to determine the etiology of the clinical alteration. Management of an intracranial hemorrhage will be performed according to each institution's usual practice.

13. Best Medical Treatment:

- The administration of medications is at the treating physician's discretion (for example intravenous fibrinolysis, anticoagulants or antiplatelet) according to local standards of care but may NOT include any intra-arterial therapies.

14. Both Arms

- Any new antithrombotic medications (usually aspirin and/or Clopidogrel) are allowed within the first 24 hours post randomization, except for patients who were treated with IV thrombolysis, until after the 24 (-6/+24) hour neuro-imaging has been performed to determine the presence/absence of intracranial hemorrhage.
- Subjects previously treated with antiplatelet agents or combination antiplatelet therapy (e.g. for a previously implanted drug eluting stent), may continue this regimen if in the investigator's opinion the benefits of continued therapy outweigh the risks of potential neurological deterioration related to hemorrhage.
- Subcutaneous Low Molecular Weight (LMW) heparin is allowed for Deep Vein Thrombosis (DVT) prophylaxis per the center's standard of care. Anticoagulants are allowed within the first 24 hours post-randomization, except for patients who were treated with IV thrombolysis, until after the 24 (-6/+24) hour neuro-imaging has been performed to determine the presence/absence of intracranial hemorrhage.
- All subjects enrolled into this study should be medically managed according to the current AHA/ASA guidelines or European Stroke Organization guidelines and, specifically as follows with regards to blood pressure and glucose management[9]. All subjects will be evaluated for standard of care treatment, including the use of thrombolytic therapy, prior to evaluation for study eligibility. Study eligibility determination will not delay standard of care stroke treatment.

- Once randomized, subjects in both arms may not be treated with any additional planned endovascular therapy or endarterectomy until after the 24 (-6/+24) hour post randomization assessments have been completed.

15. Withdrawal and Replacement of Subjects

While study withdrawal is discouraged, participant may withdraw from the study at any time, with or without reason and without prejudice to further treatment or medical care. Subjects may be also withdrawn from the study at the discretion of the PI or Sponsor for safety, or administrative reasons.

Withdrawn subjects will not undergo any additional follow-up, nor will they be replaced.

Mortality of all-cause at 90 days will be assessed for all randomized subjects. Every attempt will be made to determine the vital status of each subject who withdraws from the study, so that the withdrawal data can be used as a censoring point.

16. Follow-up Schedule per patient

All follow up time points are relative to time of randomization (time zero)

The schedule of visits/explorations (Table 1) is the same for both experimental and BMT arms subjects in the trial.

Figure 1. MOSTE Trial Flow Chart

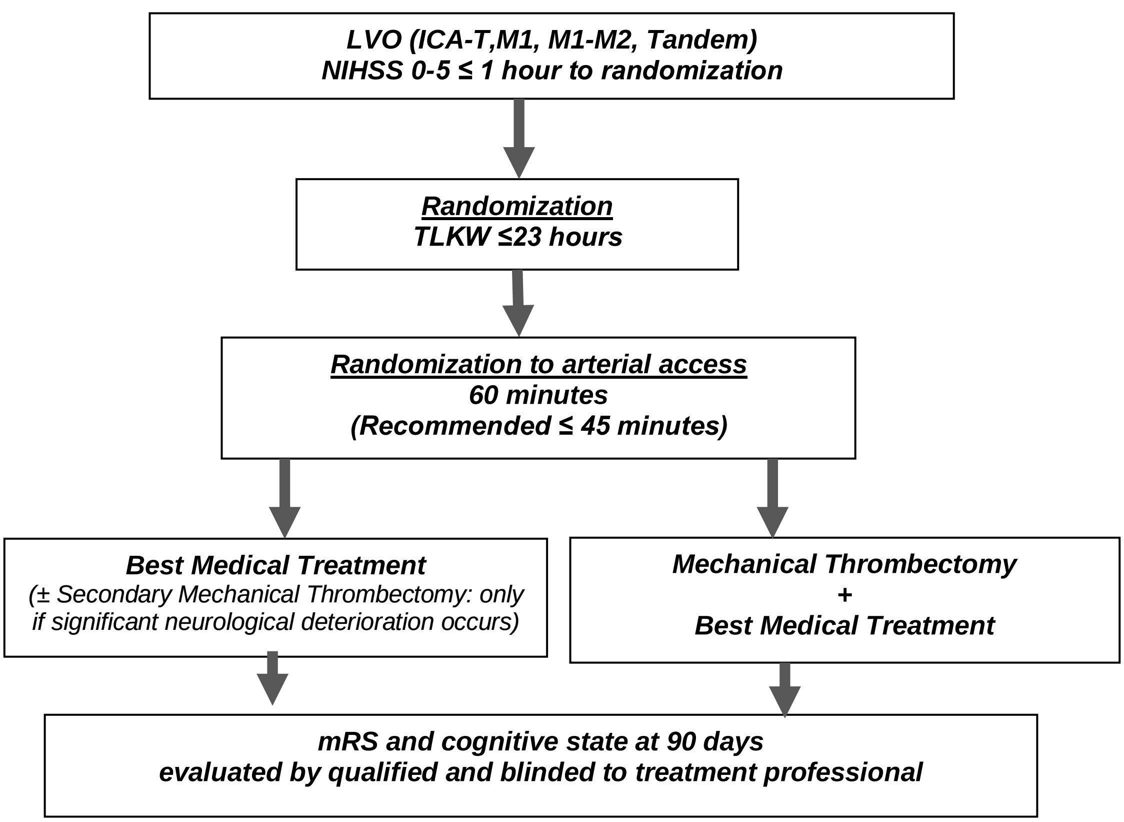

Table 1. MOSTE Trial Schedule of VISITS / EVALUATIONS

EVALUATION	Screening/ Baseline	Procedure (Treatment Arm Only)	24 H (-6/+12)[7] post randomization	Day 5-7 / Discharge (whichever is earlier)	Day 90 ± 14
Inclusion/Exclusion Criteria	✔				
Demographics / Medical History / Baseline Medications-Characteristics & Labs	✔				
Informed Consent[1]	✔				
Randomization (= time zero)	✔ [2]				
Angiography & Procedure Details [3]		✔ [4]			
mRS	✔ (pre stroke)				✔ [5]
NIHSS (all detailed items must be recorded)	✔		✔ [6]	✔	
IQ Code			✔		
EuroQol 5D/5L and QALY					✔
Montreal Cognitive Assessment (MoCA) and Trail Making tests A and B					✔
STROOP Victoria Test (French sites)					✔
NEURO IMAGING to assess for hemorrhage*** occlusion location/vessel patency, ASPECT & infarct volume (optional)	✔ MRI/MRA or CT/CTA		✔ MRI/MRA or CT/CTA	✔ (optional)	
AEs/SAEs (from time of randomization)		✔	✔	✔	✔
Concomitant Medications		✔	✔	✔	✔

(1) As a first step it can be the consent of a person of trust. In this case, the consent of the patient himself will be collected when his health conditions permit
(2) Randomization should occur within Time Last Known well ≤ 23h ;
(3) Initial CT/MR and Angiographic images will be de-identified before being submitted to core lab;
(4) For Treatment Arm only, or for Control Arm in case of secondary thrombectomy NIHSS at the time of randomization;
(5) Must be conducted by an individual assessor blinded to the randomized arm;
(6) NIHSS at 24 (-6/+12) hours: Any NIHSS relevant fluctuation will need to be recorded in the eCRF, stating its principal cause

16.1. Informed Consent

Written Informed Consent must be obtained for all subjects who are screened and meet the study eligibility criteria prior to randomization/enrollment.

The Informed Consent Form (ICF) must be approved by the study Institutional Review Board (IRB) / Ethics Committee (EC). For U.S. Sites, electronic informed consent procedures may be utilized if approved by the IRB and consistent with FDA guidance on use of electronic informed consent in clinical investigations.[1]*

The subject or the subject's Representative (acceptable patient surrogate) will be asked to sign the approved version of study Informed Consent Form before any study-specific tests or procedures are performed.

Study personnel should explain that even if a subject agrees to participate in the study and signs an Informed Consent Form, baseline non-invasive imaging or cerebral angiography may demonstrate that the subject is not a suitable candidate for the assigned study treatment and the case will be considered as screening failure.

A Screening and Enrollment Log will be maintained by the site to document basic information such as date screened and reason for SCREEN FAILURES for subjects who fail to meet the study eligibility criteria. Screen failed subjects and their reason(s) for screen failure will be documented but they will not be followed the screening visit, and no further data will be collected.

16.2. Before Randomization Procedure

The following data must be confirmed for any potential candidate before randomization and enrollment:

- Confirmation that eligibility criteria are met (all inclusion and none of the exclusion)
- NIHSS at the time of randomization. A NIHSS certified professional who is not a member of the interventional team and is recently certified in the performance of

the NIHSS will conduct the baseline NIHSS assessments to determine subject eligibility.

- Determination of NIHSS within 1 hour before randomization. Among patients transferred from primary stroke centers, admission NIHSS needs to be evaluated and recorded in the eCRF.
- MRI/MRA or CT/CTA/CTP (if MR is contra-indicated or unavailable) to assess for hemorrhage, confirms the presence of an anterior large vessel occlusion in the ICA or MCA-M1 or M1-M2 arteries with or without associated cervical lesion. This exam should be done within 1 hour before the randomization.
- ASPECT at the time of randomization

To facilitate consistency in the study schedule, the time established as zero **"t = 0"** is defined as the **time of randomization**, which occurs after initial neuro-imaging to assess for hemorrhage, confirm the presence of an anterior LVO in the anterior circulation, and after to measure the NIHSS qualifying for enrollment.

All subsequent time points (e.g. 24-hours, Day 5-7 and Day 90) will be in reference to time zero (randomization).

16.3. 24 (-6/+12) Hours post Randomization

The following data will be collected at 24 (-6/+12) hours post randomization:

- NIHSS: Any significant neurological deterioration and NIHSS relevant fluctuation during the first 24 hours will need to be recorded in the eCRF, stating its principal cause
- Neuro-imaging to determine presence/absence of hemorrhage and to assess vessel patency and infarct core volume (optional). Ideally, the same imaging modality should be used at 24 (-6/+12) hours to measure vessel patency as was used at baseline to identify occlusion location (MR imaging is recommended).

- IQCode-R (Informant Questionnaire on Cognitive Decline in the Elderly short form) : This questionnaire is completed by the family or caregivers or health care professionals to evaluate cognitive state.

16.4. Day 5-7 / Discharge

Between Day 5-7 (if subject remains in the hospital) or prior to discharge, whichever is earlier, the following data should be recorded in the appropriate eCRF panels:

- NIHSS
- Repeat imaging may be performed to re-assess final core infarct volume, at the treating physician's discretion. CT may be performed if MRI is contraindicated or not available. If performed, this imaging will be collected and reviewed by the Core Lab.

The designated staff at the investigational site will review, with the participant and prior to its hospital discharge, the study requirements in order to maximize compliance with the study follow up schedule. Participant will be reminded to return for the required follow up visits / assessments as well as to contact / inform study research staff about any adverse event or relevant change in its medical condition or changes in the prescribed treatment. It is strongly recommended to establish the date, at least for the next follow-up visit and if possible, schedule all study visits at the time of hospital discharge.

16.5. Day 90 (± 14)

On Day 90 (± 14) the following data should be recorded in the appropriate eCRF panels, within an inperson visit (by a blinded assessor):

- mRS - If participant is unable to perform an inperson visit on the Day 90 (± 14) visit, a telephone mRS assessment must be done by a qualified assessor.
- EUROQOL 5D/5L will be reported
- Montreal Cognitive Assessment (MoCA), Trail Making tests A and B, and the STROOP Victoria test (French hospitals)

Deaths MUST be reported to the CRA or designee within 24 hours of becoming aware, preferably by eCRF completion. In the event that the eCRF is unavailable, a written form sent by e-mail or fax is acceptable (the required form will be filed in the ISF). Information concerning to the primary cause of death and its date/time of occurrence will be recorded, as well as whether the subject was under consideration of “do not resuscitate” (DNR) or “comfort care only” prior to expiration.

17. Statistical Methods

17.1. Sample Size Calculation

The objective is to show the superiority of the combined best medical therapy and mechanical thrombectomy approach (experimental arm) compared to best medical therapy alone (control arm) to increase the rate of excellent functional outcome (90-day mRs 0-1, primary efficacy endpoint). Based on literature (retrospective observational studies) showing an excellent functional outcome rate ranging from 52 to 71%[15,21,23] in patients treated with best medical management, we expected a primary endpoint rate of 60% in the control arm. We assumed that this rate could be increased to 70% in the experimental arm (corresponding to an absolute increase of 10%). To detect this effect size using one-sided test (alpha risk=2.5%, power=80%), two interim efficacy/futility analysis based on O'Brien Fleming alpha and beta spending function are planned when one-third and two-thirds of the patients were recruited, and 391 patients per arm would be required (Proc seqdesign, SAS Version 9.4). To anticipate 5% of patients with missing primary endpoint (as observed in ASTER trial, see below), we planned to include 412 patients per arm (a total of 824).

To address the uncertainty concerning the excellent functional outcome rate in the control arm, the sample size will be reassessed by calculating the observed rate of excellent functional outcome in control arm at the second interim-analysis and by using the same absolute effect size (i.e. 10%). The recalculation of the sample size will be conducted by an independent statistician and submitted to the data safety monitoring board, who will decide to increase or not increase the sample size.

17.2. Loss of follow-up and Missing Data

Under the intention-to treat (ITT) principle, all patients who are randomized are included in the analysis. Every effort is to be made to minimize missing data, particularly the Day 90 outcomes.

Based on previous experiences in clinical trials of acute stroke, minimal loss to follow-up (LTFU) is expected for the 90-day assessment of the primary outcome measure. In the ASTER trial, a recent endovascular trial comparing the efficacy of two first-line mechanical strategy, the 90-day assessment of mRS was missing in 4.7% of randomized patients (n=18/381). In another recent trial comparing the efficacy of mechanical treatment following intravenous alteplase versus intravenous alteplase alone, 2.4% of randomized patients (n=10/414) had a 90-day mRS missing.

To assure that all randomized subjects will be included in the primary efficacy analysis, missing 90-day mRS values will be imputed using multiple imputation procedure on the basis of the missing at random assumption.

17.3. Statistical Analysis

Statistical analyses will be independently performed by the Biostatistics Department of University of Lille under the responsibility of Professor Alain Duhamel. For data analysis, statisticians and investigators will be aware of the experimental arm allocation. Data will be analyzed using the SAS 9.4 software (SAS Institute Inc, Cary, NC, USA). A detailed statistical analysis plan will be written and finalized prior to the database lock. Baseline characteristics will be described for each arm. Quantitative variables will be expressed as mean (standard deviation) or median (interquartile range) for non-Gaussian distribution. Qualitative variables will be expressed as frequencies and percentages. Normality of distribution will be assessed graphically and using the Shapiro-Wilk test. All analyses will be performed in all randomized patients based on their original group of randomization, according to the ITT principle. The final report will be written, based on the CONSORT statement recommendations.

The rate of patients with excellent functional outcome at 90 days after randomization will be compared between the two arms using generalized linear mixed model (GLMM) adjusted for factors considered in minimization randomization algorithm by including age (70≤ vs >70), occlusion site (ICA vs. MCA-M1 with or without cervical lesion), and symptom onset to randomization time (≤ 4.5 hours vs. >4.5 hours) as fixed effects and center as random effect. Absolute and relative risk differences with their 95% confidence interval will be derived from GLMM model as treatment effect size (experimental versus control arm). Missing values in primary endpoint will be handled by multiple imputation procedure. Imputation procedure will be perform using main baseline characteristics and treatment group. Missing values will be imputed under missing at random assumption (whatever the reason for missing data) by using regression switching approach (chained equation with m=20 imputations) with predictive mean matching method for continuous variables, logistic regression models (binary, ordinal or polynomial) for categorical variables[24]. Treatment effect estimates obtained in multiple imputed data sets will be combined using the Rubin's rules[25]. A sensitivity analysis will be conducted on the observed data (complete case analysis). In addition a sensitivity analysis will be conducted on the Per-Protocol sample defined as subjects without major protocol deviations (major deviations will be specified in the detailed statistical analysis plan).

Two interim analysis will be performed when 274 and 548 randomized patients completed the study. Decisions rules for early termination of the trial will be implemented at the interim analyses if the test statistic crosses the O'Brien Fleming boundary for efficacy (superiority) or futility, and the data safety monitoring board will consider the results of interim analyses for deciding the pursuit of trial. The statistical stopping rule at the first interim analysis is the following: if the p-value associated with the primary efficacy analysis (one-sided test) is less than 0.0001, the trial could be stopped for efficacy; if the p-value is greater than 0.594 the trial could be stopped for futility and in other case the trial could be continued. The statistical stopping rule at the second interim analysis is the following: if the p-value associated with the primary efficacy analysis (one-sided test) is less than 0.006, the trial could be stopped for efficacy; if the p-value is greater than 0.121 the trial could be stopped for futility and in other case the trial could be continued. The final analysis will be conducted with the significance level (one-sided) of 0.023 (O'Brien Fleming boundary for efficacy).

At final analysis, exploratory analyses will be performed to assess heterogeneity in treatment effect size on primary endpoint across key subgroups by including the corresponding multiplicative interaction terms in the GLMM models. From these models, treatment effect sizes (relative risk) will be estimated in each subgroup.
The following key subgroups will be investigated:

- Age (≤70 vs. >70 years)
- Site of occlusion (ICA vs. MCA-M1 or M1-M2)
- Use of IV lytics (yes vs. no, 0 – 3 hours vs > 3 hours))
- Significant neurological deficit (aphasia or weakness or hemianopia vs. others)
- Time from onset or Last Known Well to randomization (0-4.5 vs ≥ 4.5-23 hours)

Secondary binary outcomes (90-day perfect and favourable functional outcomes, 24-hours intracranial revascularization, rapid NIHSS worsening, 24-hours NIHSS 0-1, stroke in a new territory, sICH, 90-day all-cause mortality) will be also analyzed using a GLMM model including the same fixed (treatment arm, age, occlusion site and symptom onset to randomization time) and a random (center) effects; adjusted RRs will be calculated as the treatment effect size. The secondary quantitative outcome (24-hours infarct volume, 90-day QALY, MoCA, Trail Making tests A and B and the STROOP Victoria test) will be compared between the two treatments arms using a linear mixed models adjusted for factors considered in minimization randomization algorithm with a random center effect. Adjusted mean differences will be derived from linear mixed models as treatment effect sizes. In case of deviation in normality of model residuals (except if a logarithmic transformation could be applied), non-parametric analysis using Mann-Whitney U test will be used.
Among patients treated by mechanical treatment, procedure-related serious adverse events (overall and individual events rates based on subject counts and not on event counts) will be evaluated descriptively.

18. Data Collection and Management

Data will be recorded in an electronic case report forms (eCRF), developed using Clinsight (ENNOV). Clinsight is certified for the management of clinical trials and meets the recommendations of the US Food and Drug Administration. The electronic case report forms will be set up by the Biostatistics Department of University of Lille under the responsibility of Professor Alain Duhamel. All pertinent data will be entered by the study research team at investigational site into the eCRF. A unique subject ID number will be assigned to each subject. The subject will be identified only by this unique subject ID number, the first letter of the last name, the first letter of the first name and the year of birth. A list of identification of the subjects will be kept in the Trial Master File (TMF) of the investigator. The investigator will make sure that the anonymity of every person participating in the study is guaranteed.

The collected data will be integrated into a database pseudonomyzed in accordance with the French Data Protection Act. Only the medical data, the socio demographic and biological data strictly necessary to answer the objectives of the study described in the protocol can be collected and kept under the responsibility of the coordinator of the project (Pr Vincent COSTALAT).

Every reasonable effort should be made to complete data entry as soon as possible (maximum within one week) from data collection. The Principal Investigator / designee are responsible for the accuracy and completeness of recorded data.

The eCRF will be created, tested and validated before the start of data entry. The data necessary for monitoring the primary and secondary endpoints will be identified and managed at regular intervals throughout the trial. Data will be monitored by the data management team of the data-management department of University Hospital of Lille by using the predefined data management rules and queries will be automatically edited. Finally, an overall automated monitoring will be done by the data manager at the end of the data entry. In case of discrepancies, queries will be edited to resolve the problems

encountered. After validation, the database will be frozen and exported to the statistical software package for analysis.
Results from Core Labs and CEC reviews will also be entered into the EDC system and will be electronically signed by the reviewer responsible for entering this data.

The recipients of the data necessary for this study' finality are:

- Caroline Arquizan, MD, (Coordinator)
- Vincent Costalat, MD, PhD (Coordinator)
- Tudor G. Jovin, MD (Coordinator)
- Bertrand Lapergue, MD (Coordinator)
- Julien LABREUCHE (Methodologist)

19. Monitoring Procedures

Sponsor has assigned a CRO to coordinate and monitor the IN-EXTREMIS project. A separate monitoring plan will be created to describe the extent of source data verification and the details of the on-going monitoring throughout the study. Periodic monitoring visits (MV) will be made throughout the investigation to ensure that GCPs and other investigator's obligations or responsibilities are being fulfilled. These visits will be made often enough to assure that the facilities are still acceptable; the clinical investigational plan being followed; any changes to the protocol have been approved by the IRB/EC and/or reported to the sponsor and the IRB/EC; accurate, complete and timely required reports are made to Country Authorities and IRB/EC; and that all agreed-upon activities are carried out by the investigator or other specified staff members.
Original source documents will be reviewed for verification of data in the electronic database since, in order to duly perform this source data verification (SDV), the monitor must be given access to primary subject medical records. This access is disclosed to the subject via the informed consent.
It is important that the Investigator and/or relevant study site personnel are available during the MV, devoting sufficient time to the process and ensuring that it is conducted in an appropriate space.

19.1. Auditing or Inspection

Center, monitoring CRO or a regulatory agency may audit / inspect the study. By agreeing to this CIP, the investigators and their institutions accept allowance of monitoring, audits, IRB/EC review and regulatory inspections that are related to this study. They also agree to provide authorized individuals with direct access to source data and documentation, as well as the right to copy and anonymize records, provided such activities do not violate subject consent and confidentiality.

It is important that the Investigator and relevant study personnel are available during any inspection and that sufficient time is devoted to the process.

19.2. Site and Investigator selection

The Sponsor or designee (i.e. steering committee) will select investigators and site qualified by documented experience, in management of acute ischemic stroke patients and in clinical trials participation. Individual investigators will be evaluated by the Steering Committee based on experience with the intended procedure(s), and ability to conduct the clinical research according to the study protocol.

Additional selection criteria should be applied in site / investigator selection:

- ✓ Ability and willingness to randomize all eligible study subjects
- ✓ Adequate study population to meet the requirements of the study
- ✓ Adequate research staff, time and resources to support the study
- ✓ Ability to maintain treatment blinding for one or more individuals who will collect mRS data at 90 days f-up. Experience in conducting randomized, controlled, clinical studies
- ✓ Willingness to observe required confidentiality
- ✓ Adherence to protocol and willingness to provide the Study with accurate data in a timely manner.
- ✓ Association with an IRB / EC that satisfies all applicable regulatory requirements

19.3. Investigator Responsibilities

Each research team member (authorized by site PI in the Authorization Signature Log) is responsible for conducting this study in accordance with the protocol, Good Clinical Practice, the Investigator agreement, all applicable laws and regulations, and any conditions of approval imposed by the reviewing IRB/EC or CA.

19.4. Training of Investigator and Site Personnel

The training of the Investigator, and appropriate clinical site personnel will be a shared responsibility between Sponsor and CRO and may be conducted during an investigator meeting, a site initiation visit, or other appropriate training sessions.

Training will include, but not be limited to, the study protocol, CRF completion and site personnel responsibilities. All Investigators and site personnel that are trained must have signed the Study Authorization Signature Log.
Each investigator (neurologist and neuroradiologist) has to be certified for NIHSS, mRS and GCP (via electronic forms). A NIHSS certified professional who is not a member of the interventional team and is recently certified in the performance of the NIHSS will conduct the baseline NIHSS assessments to determine subject eligibility.
The blinded evaluator of mRS has to be certified for mRS (via electronic forms).

20. Adverse Events management

Regarding the vigilance of the project, the responsibilities of the investigator and sponsor, the reporting of serious adverse events, and annual safety reports will be monitored and carried out in accordance with local regulations.

Concerning the practical modalities of trial vigilance in Spain, please refer to Appendix E.

Below, the practical modalities of trial vigilance in France.

20.1. Adverse Event Definitions and Classification

<u>Adverse event</u> (article R1123-46 of the French public health code)
Any harmful event occurring in a person who is enrolled in a research involving the human person, whether or not linked to the research or product to which the research relates.

<u>Adverse reaction</u> (article R1123-46 of the French public health code)

An adverse event occurring in a person who is enrolled in a research involving the human person, when that event is related to the research or product to which the research relates.

Serious Adverse event or reaction (article R1123-46 of the French public health code and ICH E2B guide)

Any adverse event or reaction that:

- ✓ causes death,
- ✓ endangers the life of the person who is enrolled in a research,
- ✓ requires hospitalization or prolongation of hospitalization,
- ✓ causes a significant or persistent disability or incapacity,
- ✓ results in a congenital anomaly or malformation,
- ✓ or any event considered medically serious,

And for the drug, regardless of the dose administered.

The expression “life-threatening” is reserved for an immediate life threat at the time of the adverse event.

Note: Planned hospitalization for pre-existing condition, or a procedure required by the Clinical Investigation Plan, without a serious deterioration in health, is not considered a serious adverse event.

Adverse reaction unexpected (article R1123-46 of the French public health code)

Any adverse reaction whose nature, severity or development is inconsistent with information on the products, practices and methods used during the research.

New fact (article R1123-46 of the French public health code)
Any new data which may lead to a reassessment of the relation between the benefits and the risks of the research or product of the research, to changes in the use of this product, in the lead of the research, or documents of the research, or to suspend, interrupt, modify the protocol of the research.

20.2. Serious and not serious expected adverse reaction

Reference safety document: The current protocol, manufacturer device notice, summaries of products characteristics will be considered as the reference safety document.

The adverse effects expected (not exhaustive) with this protocol are those related to:

- Mechanical thrombectomy:

- Vascular perforation
- Arterial dissection
- Embolization to new territory
- Access site complication requiring surgical repair or blood transfusion
- Intra-procedural mortality
- Device failure (in vivo breakage)
- Any other complications adjudicated by the CEC to be related to the procedure

- MRI/MRA: Dizziness, metallic taste in the mouth, changes in heart rate, muscle fibrillation, burns

- CT/CTA: X-ray exposure. However, the doses of the rays used are weak and therefore do not entail undesirable effects.

- Medical treatments: All adverse reactions cited in the summaries of products characteristics.

20.3. Events of interest

The following conditions are events of interest and should be recorded in any case, as serious adverse events:

- Symptomatic intracerebral hemorrhage
- Neurological status deterioration (≥4 points increase in NIHSS score or significant according to the investigator)
- Worsening of stroke
- Recurrent stroke in the same territory
- Recurrent stroke in a new territory

The Procedure specific adverse events are also events of interest and must be recorded in any case as serious adverse events. The following events are Procedure specific adverse events:

- Vessel perforation
- Intracranial artery dissection
- Embolism of a new territory
- Vascular access site complication
- Device malfunction
- Contrast media reaction
- Cerebral vasospasm

20.4. Investigator responsibilities

Collection of AEs

Complete and appropriate data on all AEs experienced during the clinical trial will be recorded on the AE form of the CRF on an ongoing basis for the duration of the study. Each AE report shall include a description of the event, an assessment of its seriousness according to the criteria listed above, its duration, intensity, relationship to the study devices, other causality factors (if any), any concomitant medication dispensed, actions taken with the study drug or other therapeutic interventions and outcome at the end of the observation period.

For each AE, a separate AE form will be filled in.
AEs will be followed up until their resolution or stabilisation. However, the observation period will be cut off after the last patient has finished his final visit (LPLV).

All AEs and the treatment and follow-up required must be documented in the subject's medical records and in the eCRF.

Exception to collection

The following circumstances will not be collected:

- ✓ Admission for social or administrative reason
- ✓ Hospitalization predefined by the protocol
- ✓ Hospitalization for medical or surgical treatment planned before the research

Notification of SAEs

The investigator evaluates each adverse event in terms of severity.
The investigator must notify to the Sponsor all serious adverse events, without delay, the day the event is known, if it occurs:

- ✓ At the **time of enrollment** (defined as **time of randomization**)
- ✓ During all the duration of the patient participation

If the investigator becomes aware of a serious adverse event, which he suspects a causal relationship to the research after the end of the clinical trial in a participant, he must inform the Sponsor without delay.
The investigator must document the event as well as possible, and give the medical diagnosis if possible**.** The investigator should ensure the relevant follow up information is communicated to the Sponsor as soon as possible.
The investigator has to provide with the serious adverse event notification form, the copies of results or hospitalization reports relating the serious adverse event. These documents must be anonymous and the number and code of the patient in the study must be subscribed.
The investigator has to follow the patient which had a serious adverse event until the resolution of the event, even if the follow up of the study is completed**.** A complementary information about the event evolution will be sent by the investigator to the Sponsor.
The investigator and the Sponsor have to evaluate, independently, the causal link between the serious adverse event, the treatments and the study.
All serious adverse events for which the investigator or the Sponsor believes a causal relationship can be reasonably considered are considered suspected serious adverse reactions.
The investigator must notify every new fact that come to his knowledge to the sponsor.
Moreover, the events concerning a concomitant treatment must be reported to the regional pharmacovigilance centre on which the investigator depends geographically.

Regarding subject deaths, a copy of the autopsy report needs to be sent to the sponsor's designee, when applicable and available. Any other source documents related to the death should also be provided. In the event that no source documents are available, the PI or delegate will make all possible efforts to obtain as much information as possible (i.e. contacting relatives or other medical institutions) and he/she is required to describe the circumstances of the subject's death in a letter, e-mail or other written communication. Death should not be recorded as an adverse event, but should only be reflected as an outcome to another specific AE/SAE.

Exception to notification

Some serious adverse events are not requiring immediate reporting but only be documented on the CRF/ source data as AE.

Underlying (pre-existing) symptoms or diseases are not reported as Adverse Events (AEs) unless there is an increase in severity or frequency during the course of the investigation, but they need to be reported in the eCRF as Relevant Medical History

Summary table of the notification circuit by type of event

Sort of event	Notification procedures	Period of the notification to the Sponsor
Non serious adverse event	In the CRF	No notification
Serious adverse event or event of interest	Serious adverse event declaration form and follow up if necessary + Report in the CRF	**Notification without delay to the Sponsor**
New fact	Written report	**Notification without delay at the Sponsor**

In case of notification, the declaration has to be made with a serious adverse event declaration form and send by fax or email at:

Direction de la recherche et de l'Innovation
Vigilance des Essais Cliniques
Perrine ROBIN
Fax : +33 4 67 33 54 11

Vigilance-ec@chu-montpellier.fr

This initial notification is a written report and must be followed, if necessary, by detailed written supplementary report(s).

20.5. Sponsor responsibilities

Assessment of Serious Adverse Events

The assessment of the relationship between a SAE and the administration of experimental procedure is a clinical decision based on all available information at the time of, and after the occurrence of the event. This assessment has to be done by sponsor too.

In case of positive relationship between the event and the experimental procedure, the sponsor will determine whether the events are expected or unexpected based on the reference safety document.

Expedited reporting

Other safety issues also qualify for expedited reporting where they might materially alter the current benefit-risk assessment of the investigational devices or that would be sufficient to consider changes in the investigational devices administration or in the overall conduct of the trial, for instance:

a) An increase in the rate of occurrence or a qualitative change of an expected serious adverse reaction (SESAR), which is judged to be clinically important,

b) Post-study SUSARs that occur after the patient has completed a clinical trial and are reported by the investigator to the sponsor,

c) New events related to the conduct of the trial or the development of the investigational devices and likely to affect the safety of the subjects, such as:

- A SAE which could be associated with the trial procedures and which could modify the conduct of the trial,
- A major safety finding from a newly completed animal study (such as carcinogenicity)

Declaration of Suspected Unexpected Serious Adverse Reaction (SUSAR)

The sponsor reports, in accordance to the regulatory period, the serious and unexpected adverse reactions to the competent authorities and inform investigators.
In France, the reglementary declaration has to be made in a maximal period of:

- Without delay for serious and unexpected adverse reaction fatal or life-threatening. In this case, additional relevant information must be sought and transmitted within a maximum of 8 days.
- 15 calendar days for all serious and unexpected adverse reaction. Additional relevant information must be sought and transmitted within a period of 8 days.

The sponsor shall promptly declare any new facts that occurred in the research:

- to the ANSM,
- the Committee for the Protection of Persons

The sponsor and the investigator shall take appropriate urgent measures. The sponsor shall inform the competent authorities and the Committee for the Protection of Persons.

Annual safety report (ASR)

In addition to the expedited reporting, the sponsor shall submit, once a year, throughout the clinical trial or on request, a safety report to the competent authorities and the Ethics Committee of the concerned Member States taking into account all new available safety information received during the reporting period.

The aim of the ASR is to assess the safety conditions of subjects included in the concerned trial(s).

It should be the same for the competent authorities concerned and the Ethics Committee concerned.

The Sponsor is responsible for the ongoing safety evaluation of the Investigational Procedure. The Sponsor should promptly notify all concerned investigators and the regulatory authorities of findings that could affect adversely the safety of subjects, impact the conduct of the trial, or alter the CA approval/ EC's favourable opinion to continue the trial.

Data Safety monitoring Committee

The DSMC or DMC (Data Monitoring Committee) is an advisory committee that provides guidance to the sponsor with respect to the benefit - risk balance and the conduct of the clinical trial.

Its members are not involved in the study and are chosen for their competence in one or more of the investigative fields of the study in collaboration with the coordinating investigators.

The DMC is composed of at least:

· a clinician specialized in the pathology studied to evaluate the clinical aspects
of tolerance and efficacy of the study

· a methodologist or statistician.

The opinion of the DMC will be sent to the sponsor in a timely manner. The sponsor will forward this opinion to the coordinating investigator, the Ethic Committee and the CA as part of the safety annual report.

21. Risk Benefit Analysis

It is possible that subjects enrolled into this trial will receive no direct benefit from participation. There may be additional risks to subjects randomized to the mechanical thrombectomy plus medical management arm in addition to those that are currently known or anticipated for patients presenting with a Large Vessel Occlusion Stroke with Minor Symptoms (NIHSS <6).

Due to differences in standards of care between sites, it is possible that some subjects may receive additional follow-up imaging or neurologic examinations other than those required by the protocol. The risks of these neurologic examinations are negligible, and the subject would likely benefit from enhanced care due to these additional tests.

Only trained and experienced physicians will perform EVT within the trial, accredited by the study form “Annex to CV – Neuro-Interventionism Statement of Experience”.

Since the patients in whom the mechanical thrombectomy will be performed within this study will present with a Large Vessel Occlusion Stroke with Minor Symptoms (NIHSS <6), participation in the study adds a currently unknown level of risk to the subjects who are randomized to the Treatment Arm. However no publications have confirmed this finding, and the potential benefits of mechanical thrombectomy plus medical management include higher revascularization rates, and revascularization has been shown to be correlated with improved clinical outcomes and reduced mortality[26], therefore potential benefits outweigh the anticipated risks.

The main potential risks in the treatment arm receiving immediate EVT as first line therapy consists in an additional risk related to endovascular procedure such as: Ischemic lesions due to collateral emboli during endovascular treatment, bleeding risk due to wire perforation during micro-catheterization of the brain vessels, and groin puncture complications such as infection and pseudo-aneurysm. There is also risk due to the radiation associated with the use of fluoroscopy; this risk includes radiation-induced injuries to the skin and underlying tissues ("burns"), which occur shortly after the exposure, and radiation-induced cancers, which may occur later in life.
The main potential risks in the control arm receiving only the medical treatment and EVT in secondary theraphy is to suffer from a delayed reperfusion resulting in an increased stroke volume, and therefore a worse outcome (33% mrs 0-2)22, 23. The EVT complications overall are expected to be as low as previously reported in the literature, that is under 6%[15, 16, 17] making the risk/benefit ratio in favor of the therapeutic arm offering immediate and full recanalization to the patient.
If reperfusion has <u>not</u> been successful with the chosen retriever (defined as modified **TICI 0-2a** in the treated territory) continue with additional retrieval attempts (the number of passes is up to the interventionist decision but it is strongly recommended to not exceed the 3 passes per device). Additional therapy may be initiated after the 3 passes if deemed appropriate by the treating physician, but it is discouraged as at that point it will be more harmful than beneficial to continue.

21.1. Risk Minimization

Each site must obtain IRB or EC approval prior to screening and enrolling subjects.

Every subject or Representative will be required to provide signed Informed Consent prior to participation which will explain their treatment choices and the risks and benefits of being in the study.

Finally, several committees and core labs will assist in oversight of the study which ensures that any risks to subjects will be minimized. Additionally, risk will be mitigated in the mechanical thrombectomy plus medical management arm by implementation of an adaptive trial design which allows for early and frequent assessment of efficacy and safety parameters in the two study arms, to ensure that the number of subjects exposed to a potentially non-beneficial treatment is minimized.

Safety monitoring of the data, consisting of individual event and aggregate data review, will be ongoing and conducted at a rate commensurate with subject enrollment in the trial.

22. Study Committees, Core Labs and Partner

22.1. Steering Committee

A Steering Committee has been convened. Responsibilities include oversight of the overall conduct of the study with regard to protocol review and development, study progress, ensuring adequate subject safety oversight, and overall data quality and integrity.
The Steering Committee will oversee dissemination of study results through appropriate scientific sessions and publications. The Steering Committee may select additional investigators, based on enrollment and adherence to the protocol, to participate on a **Publication Committee**, which will participate in the review and approval of all requests for data analysis, abstract and manuscript preparation and submission.

22.1.1 Publication rules

Final report

The analysis of the data will be done by the methodologist of the study.

A final report dated and signed by the coordinating investigators will be forwarded to the sponsor, who will forward it to the competent authorities within 12 months of the end of the study.

Publication rules

Any written or oral communication of the research results must receive the prior agreement of the coordinating investigators and the sponsor.
The University Hospital of Montpellier is the owner of the data and no use or transmission to a third party can be carried out without its prior agreement.
The University Hospital of Montpellier must be mentioned as the sponsor of the research.
Model of the address of the University Hospital of Montpellier in the publications:
CHU Montpellier, department / service, city, F-postal code, country.

Communication of results to participants

According to the applicable local country laws, the subjects are informed, at their request, of the overall results of the research by the investigator.

Transfer of data

The conditions for the transfer of all or part of the research database are decided by the research sponsor and are the subject of a written contract.

22.2 Safety Monitoring Committees

To promote early detection of safety issues routine medical monitoring will be conducted on an ongoing basis. In addition, the CEC charters will provide for evaluation of safety events at routine intervals.
The CEC may be un-blinded due to the fact one study group receives an intervention while the second study group does not. The dataset will contain obvious AEs/SAEs specific to the interventional procedure that will, simply by their presence, un-blind those individuals reviewing the data.
This process requires the dynamic collection of unmonitored data as soon as an event is reported. This is expedited by designated Sponsor Safety Personnel, who are responsible

for reviewing safety data within the trial on an ongoing basis, and coordinating the collection of information for inclusion within the CEC event dossier, from the sites and Core Labs. During regularly scheduled monitoring visits, the clinical research monitors will support the dynamic reporting process through their review of source document information.

22. 3 Clinical Events Committee (CEC)

The CEC will include specialists in stroke (neurology and neuro-interventionalist) as well as other experts with the necessary therapeutic and subject matter expertise, who are not participating in the trial.

CEC responsibilities, qualifications, membership, and committee procedures are outlined in the specific CEC charter.

The CEC will be responsible to review and adjudicate the following protocol-defined safety outcome measures and relevant adverse events reported by study investigators.

Relevant information and source documents will be provided to assist with their review and adjudication, for the following events:

- Events of interest
- All neurological SAE
- All deaths

22.4 Imaging Core Labs

Independent and Centralized Imaging Core Lab, by University of California, Los Angeles (UCLA - 635 Charles E Young Drive South, Suite 225 - Los Angeles, CA 90095-7334), will be used in this study to provide consistent, independent evaluation of images for confirmation of inclusion criteria. The OLEA Company (93 avenue des Sorbiers, ZI Athélia IV, 13600 La Ciotat)is in charge of the harmonization and the recovery of the images to the transfer to the CoreLab.

The imaging core lab will review CT/MR images obtained at baseline and at 24 (-6/+12) hours post randomization to determine vessel patency, hemorrhage, and extent of infarct.

The imaging core lab will also review angiographic images from the procedure, to determine revascularization and clot location.
Ideally MR imaging will be used whenever possible to screen subjects for inclusion into the trial. However, if MR imaging is contraindicated or is unavailable then CT based imaging may be utilized.
Ideally the same imaging modality used at baseline will be used at 24 (-6/+12) hours post randomization. However, for subjects with compromised renal function who had a CT/CTA at baseline, but in whom the treating physician wishes to avoid an additional load of contrast, an MRI/time-of-flight MRA of the intracranial arteries may be obtained instead.

An imaging **core lab charter** will ensure that consistent policies and procedures are applied throughout the imaging core lab review and determination process. Sites will be provided with this charter prior to study initiation visit, which will include instructions for how images should be collected and submitted to core lab within 2 weeks of acquisition of the final required imaging time point at 24 (-6/+12) hours after the procedure. If this timeline cannot be met for any reason, the site should communicate this delay to the CRA or designee so that the pending images can be tracked until received.
Core lab will send periodically to CRA or designee the list of images received, after performing basic verification of the images received, will archive all images and forward the applicable results to the CEC or designee when applicable.

The CT/MR Core Lab will verify, and record, the ASPECTS and will also provide timely feedback to the study sites regarding quality control issues.

Angiograms must be appropriately de-identified, before being sent to the imaging core lab for evaluation. Subject identifying information includes name, medical record number, social security number, as well as address and telephone number. Subject’s date of birth will be required in order to identify images provided from the study sites.

It is important that the images be saved in native DICOM format, and that all imaging sequences are sent (without pre-selecting specific frames). It is also important that the imaging sequences are captured chronologically and are clearly labeled with date and time stamps so that they can be correlated to pre-procedure, post-retriever, and post-procedure time points. Specific imaging transmittal instructions will be provided to the sites by the imaging core lab (proposed use of DICOM Grid / Ambra Health cloud-based, medical image management suite)

22.5 A.R.N.G.D.C

The Association for Research in Neuroimaging of Gui De Chauliac (A.R.N.G.D.C) contributes to:

- The development and promotion of research in neurological, ophthalmological, otorhinolaryngological, medical, surgical and interventional imaging.
- The development of all actions having for object the training of the medical and technical personnel in the sector of neuroradiology.

A.R.N.G.D.C is managed by a board of 3 members, whose president is Pr Vincent COSTALAT and the head office is located at the Neuroradiology Department of the Hôpital Gui de Chauliac-CHU de Montpellier, 80 Avenue Augustin Fliche, 34295 MONTPELLIER CEDEX 5.

The commitments of the A.R.N.G.D.C in the framework of the two projects MOSTE and LASTE are the following:

- Scientific expertise in neuroradiology and neurology for the drafting of research protocols and amendments
- Collaboration in the coordination and management of the projects
- Collaboration in logistical management
- Development of the IN EXTREMIS study website https://www.inextremis-study.com/

23. Ethical Considerations

23.1 Compliance with Good Clinical Practices (GCP) and Adherence to Protocol

The Investigator will ensure that this study is conducted in accordance with the European regulations *Regulation (EU) 2017/745 of the European Parliament and of the Council of 5*

April 2017 on medical devices, amending Directive 2001/83/EC, Regulation (EC) No 178/2002 and Regulation (EC) No 1223/2009 and repealing Council Directives 90/385/EEC and 93/42/EEC the ethical principles that have their origins in the Declaration of Helsinki and that are consistent with GCP and applicable regulatory (local) requirements; whichever affords the greater protection to the subject.

Prior to beginning the study, the Investigator must sign the Investigator Agreement and Signature page documenting his/her agreement to conduct the study in accordance with the current version of study protocol or clinical investigational plan (CIP).

An Investigator must not make any changes or deviate from this protocol, except to protect the life and physical well-being of a subject in an emergency.

Any deviation from the protocol must be documented with the date and reason for the deviation and reported to Sponsor as soon as possible (if relevant enough) or through the monitoring report. Protocol deviations will be communicated to the IRB/EC or to CA, if relevant and as per local guidelines and government regulations.

Major and minor protocol deviations will be defined / classified by the Study Steering Committee previous to any data analysis performance.

23.2 Institutional Review Board/ Ethics Committee

Final version of the protocol/CIP, proposed Patient Information Sheet and Informed Consent Form (PIS+ICF), other written subject information, as well as any proposed advertising material must be submitted to the IRB/EC for written approval. A copy of the written IRB/EC approval of the protocol and PIS+ICF, stating date and versions, must be received by Sponsor or designee before start of subjects recruitment into the study.

Any further protocol amendment and changes to the PIS + ICF should be submitted for approval to the IRB/EC or CA as applicable by in force regulations.

The Investigator is responsible for obtaining annual IRB/EC approval and renewal throughout the duration of the study where required.

Sponsor or designee will obtain approval of PIS+ICF by each pertinent EC or RA, for use at his/her site. The PIS and ICF documents should be translated into the language(s) understandable to potential subject population(s) and also adapted to the local requirements. Sponsor and the reviewing IRB/EC must approve the ICF before use at each participating site. The ICF must be in agreement with current Good Clinical Practices (GCP) guidelines, the Declaration of Helsinki, and the International Conference on Harmonization (ICH) as well as any other applicable local regulation.

Before participating in the clinical trial, each subject or his/her representative (or acceptable patient surrogate) must give written Informed Consent, after the context of the study has been fully explained in a language that is easily understood by the subject or its representative. The subject or representative must also be given the opportunity to ask questions and have those questions answered to his/her satisfaction.

Written Informed Consent must be recorded appropriately by means of the subject's, or representative's dated signature. The consent process must be documented in the subject's medical chart. At U.S. sites, electronic informed consent may be utilized in accordance with FDA's Guidance on the Use of Electronic Informed Consent in Clinical Investigations if approved by the site's IRB.

Note - If approved by local ethics committee and country regulations, the investigator is allowed to enroll a patient if, the subject or the representative or person of trust is not available to sign. However, as soon as possible, the patient is informed and his/her consent is requested for the possible continuation of this research (not applicable to U.S. Sites)

23.3 Insurance

The CHU of Montpellier, sponsor of the study, signs for all the study duration an insurance guaranteeing its own civil liability as well as that of every participant involved in the study, independently of the nature of the links existing between the participants and the Sponsor.

24. Study Administration

24.1. Pre-Study Documentation Requirements

Prior to enrolling any subjects into the trial the site must complete all pre-study essential documents, and these must be confirmed to be on file:

Documentation of a completed Site Initiation Visit

24.2. Record Retention

The Investigator will maintain all essential trial documents and source documentation, in original format, that support the data collected on the study subjects in compliance with the ICH/GCP guidelines and local requirements.

Study documents must be retained until at least 2 years have elapsed since the formal discontinuation of the clinical investigation of the product. These documents will be retained for a longer period of time in compliance with in force local regulatory requirements. The Investigator will take measures to ensure that these essential documents are not accidentally damaged or destroyed. If for any reason the Investigator withdraws responsibility for maintaining these essential documents, custody must be transferred to an individual who will assume responsibility. Sponsor must receive written notification of this custodial change.

24.3. Criteria for Terminating Study

Sponsor reserves the right to terminate the study but intends only to exercise this right for valid scientific or administrative reasons and reasons related to protection of subjects. Investigators and associated IRB/EC will be notified in writing in the event of termination.

24.4. Criteria for Suspending/Terminating a Study Site

Sponsor reserves the right to stop the enrollment of subjects at a study site at any time after the study initiation visit if no subjects have been enrolled or if the site has multiple or major protocol violations without justification or fails to follow remedial actions. Notification of termination of a Study Site will be made by Sponsor or designee to the appropriate regulatory agencies and IRB/EC, as required.

25. Recommendations

Mechanical Thrombectomy Procedure

In subjects randomized to the mechanical thrombectomy plus medical management arm, the procedure should be performed using upon neuro-interventionist decision, the stent retrievers used as per regular practice at investigational site. The devices will be selected from the list of devices approved for use in the study (CE labeled or FDA cleared for revascularization with NRY code).

NOTE: Although the protocol allows up to 60 minutes between randomization and arterial access it is strongly recommended to start the procedure within 45 minutes after randomization.
The interventional procedure should be completed within two (2) hours of arterial access.

Heparin anticoagulation may be used but should not exceed a total of 3,000 units of Heparin bolus followed by 500 units/hour Heparin drip for the duration of the procedure.
Prudent use of anti-vasospasm agents is permitted.
Use of the stent retriever should be terminated if there is any angiographic evidence leading to the suspicion of an intracranial hemorrhage, such as extravasation of contrast during the procedure.
Physicians should follow the most current Instructions for Use (IFU) at all times with regards to the used device compatibility, preparation and the recommended retrieval procedure.

Immediately after each retrieval attempt with the chosen retriever, perform biplane angiography in order to assess the vessel patency in the neurovascular tree that is being treated. Angiography should include ipsilateral AP and lateral imaging of the involved arterial system, including potential collateral vessels.

If reperfusion has been successful with the chosen retriever (defined as at least **TICI 2b** (reperfusion of > 50% MCA territory) in the territory treated) the mechanical thrombectomy procedure should be stopped and no further interventions performed.

If reperfusion has not been successful with the chosen retriever (defined as modified **TICI 0-2a** in the territory treated) continue with additional retrieval attempts (the number of passes is up to the interventionist decision but it is strongly recommended to not exceed the 3 passes per device). Additional therapy may be initiated after the 3 passes if deemed appropriate by the treating physician, but it is discouraged as at that point it will be more harmful than beneficial to continue.

Hypertension management / BP considerations

The peri-stroke management of out of range values of arterial BP remains controversial.
In subjects who received IV tPA, blood pressure should be managed according to post IV tPA management protocols (systolic blood pressure is <180 mm Hg and their diastolic blood pressure is <105 mm Hg) within the first 24 hours.
In subjects who are re-perfused after mechanical embolectomy (defined as achieving TICI 2b or TICI 3), systolic blood pressure is encouraged to be maintained below 140 mm Hg in the first 24 hours to minimize the risk of reperfusion related ICH. In subjects who do not achieve recanalization after thrombectomy, similar B/P management orders should be applied as for the control subjects within each center.
It is observed that many patients have spontaneous declines in blood pressure during the first 24 hours after onset of stroke, and it is medically consensual that a cautious approach to the treatment of arterial hypertension should be recommended.
Due to the fact that arterial BP is a dynamic parameter, it is important to monitor it frequently, especially during the first day of stroke, to identify trends and extreme fluctuations that would require intervention.

Perfusion data collection
If CT perfusion or PWI-MRI are performed during routine stroke management, the data should be collected and documented in the e-CRF in the “Complementary Imaging Data” section by the performing center. This data will be used for ancillary analysis.

26. References

1. WHO. Burden of Stroke. 2017. http://strokeeurope.eu/burden-of-stroke-report-launched-in-eu-parliament/.
2. Smith WS, Tsao JW, Billings ME, et al. Prognostic significance of angiographically confirmed large vessel intracranial occlusion in patients presenting with acute brain ischemia. Neurocrit Care. 2006;4(1):14-17. http://www.ncbi.nlm.nih.gov/pubmed/16498189. Accessed September 1, 2017.
3. Heidenreich PA, Trogdon JG, Khavjou OA, et al. Forecasting the Future of Cardiovascular Disease in the United States. Circulation. 2011;123(8). http://circ.ahajournals.org/content/123/8/933.long. Accessed September 1, 2017.
4. Reeves M, Khoury J, Alwell K, et al. Distribution of National Institutes of Health stroke scale in the Cincinnati/Northern Kentucky Stroke Study. Stroke. 2013;44(11):3211-3213. doi:10.1161/STROKEAHA.113.002881.
5. Dhamoon MS, Moon YP, Paik MC, et al. Long-term functional recovery after first ischemic stroke: The Northern Manhattan Study. Am J Epidemiol. 2009;40(8):2805-2811. doi:10.1161/STROKEAHA.109.549576.Long-term.
6. Romano JG, Smith EE, Liang L, Gardener H, Camp S, Shuey L, Cook A, Campo-Bustillo I, Khatri P, Bhatt DL, Fonarow GC, Sacco RL SL. Outcomes in mild acute ischemic stroke treated with intravenous thrombolysis: a retrospective analysis of the Get With the Guidelines-Stroke registry. JAMA Neurol. 2015;72(4):423-431. doi:10.1001/jamaneurol.2014.4354.
7. Ali SF, Siddiqui K, Ay H, Silverman S, Singhal A, Viswanathan A, Rost N, Lev M SL. Baseline Predictors of Poor Outcome in Patients Too Good to Treat With Intravenous Thrombolysis. Stroke. 2016;47(12):2986-2992.
8. Goyal M, Menon BK, van Zwam WH, et al. Endovascular thrombectomy after large-vessel ischaemic stroke: a meta-analysis of individual patient data from five randomised trials. Lancet. 2016;387(10029):1723-1731. doi:10.1016/S0140-6736(16)00163-X.
9. Powers WJ, Derdeyn CP, Biller J, et al. 2015 American Heart Association/American Stroke Association Focused Update of the 2013 Guidelines for the Early Management of Patients With Acute Ischemic Stroke Regarding Endovascular Treatment. Stroke. 2015;46(10):3020-3035. doi:10.1161/STR.0000000000000074.
10. Maas MB, Furie KL, Lev MH, et al. National Institutes of Health Stroke Scale score is poorly predictive of proximal occlusion in acute cerebral ischemia. Stroke. 2009;40(9):2988-2993. doi:10.1161/STROKEAHA.109.555664.
11. Laurencin C, Philippeau F, Blanc-Lasserre K, Vallet AE, Cakmak S, Mechtouff L, Cho TH, Ritzenthaler T, Flocard E, Bischoff M, El Khoury C, Nighoghossian N DL. Thrombolysis for Acute Minor Stroke: Outcome and Barriers to Management. Results from the RESUVAL Stroke Network. Cerebrovasc Dis. 2015;40(1-2):3-9.
12. Rajajee V, Kidwell C, Starkman S, Ovbiagele B, Alger JR, Villablanca P, Vinuela F, Duckwiler G, Jahan R, Fredieu A, Suzuki S SJ. Early MRI and outcomes of untreated patients with mild or improving ischemic stroke. Neurology. 2006;67(6):980-984.
13. Jovin TG, Albers GW, Liebeskind DS, STAIR IX Consortium. Stroke Treatment Academic Industry Roundtable. Stroke. 2016;47(10):2656-2665. doi:10.1161/STROKEAHA.116.013578.
14. Goyal M, Yu AY, Menon BK, Dippel DW, Hacke W, Davis SM, Fisher M, Yavagal DR, Turjman F, Ross J, Yoshimura S, Miao Z, Bhatia R, Almekhlafi M, Murayama Y, Sohn SI,

Saver JL, Demchuk AM HM. Endovascular Therapy in Acute Ischemic Stroke: Challenges and Transition From Trials to Bedside. Stroke. 2016;47(2):548-553.
15. Dargazanli C, Consoli A, Gory B, et al. Is Reperfusion Useful in Ischaemic Stroke Patients Presenting with a Low National Institutes of Health Stroke Scale and a Proximal Large Vessel Occlusion of the Anterior Circulation? Cerebrovasc Dis. 2017;43(5-6):305-312. doi:10.1159/000468995.
16. Park TH, Hong K-S, Choi JC, et al. Validation of minor stroke definitions for thrombolysis decision making. J Stroke Cerebrovasc Dis. 2013;22(4):482-490. doi:10.1016/j.jstrokecerebrovasdis.2013.03.006.
17. Fischer U, Baumgartner A, Arnold M, et al. What is a minor stroke? Stroke. 2010;41(4):661-666. doi:10.1161/STROKEAHA.109.572883.
18. Briggs DE, Felberg RA, Malkoff MD, Bratina P, Grotta JC. Should mild or moderate stroke patients be admitted to an intensive care unit? Stroke. 2001;32(4):871-876.
19. Pfaff J, Herweh C, Pham M, et al. Mechanical Thrombectomy in Patients with Acute Ischemic Stroke and Lower NIHSS Scores: Recanalization Rates, Periprocedural Complications, and Clinical Outcome. AJNR Am J Neuroradiol. June 2016. doi:10.3174/ajnr.A4862.
20. Haussen DC, Bouslama M, Grossberg JA, et al. Too good to intervene? Thrombectomy for large vessel occlusion strokes with minimal symptoms: an intention-to-treat analysis. J Neurointerv Surg. 2017;9(10):917-921. doi:10.1136/neurintsurg-2016-012633.
21. Haussen DC, Lima FO, Bouslama M, et al. Thrombectomy versus medical management for large vessel occlusion strokes with minimal symptoms: an analysis from STOPStroke and GESTOR cohorts. J Neurointerv Surg. August 2017. doi:10.1136/neurintsurg-2017-013243.
22. Messer MP, Schonenberger S, Mohlenbruch MA, et al. Minor Stroke Syndromes in Large-Vessel Occlusions: Mechanical Thrombectomy or Thrombolysis Only? AJNR Am J Neuroradiol. 2017;38(6):1177-1179. doi:10.3174/ajnr.A5164.
23. Dargazanli C, Arquizan C, Gory B et al. Mechanical Thrombectomy for Minor and Mild Stroke patients harboring Large Vessel Occlusion: A Multicenter Case Control Study. Stroke. 2017;
24. Buuren S van, Groothuis-Oudshoorn K. mice: Multivariate Imputation by Chained Equations in R. J Stat Softw. 2011;45(3). doi:10.18637/jss.v045.i03.
25. Rubin DB. Introduction, in Multiple Imputation for Nonresponse in Surveys. Hoboken, NJ, USA.: John Wiley & Sons, Inc.; 1987. doi:10.1002/9780470316696.ch1.
26. Rha J-H, Saver JL. The Impact of Recanalization on Ischemic Stroke Outcome: A Meta-Analysis. Stroke. 2007;38(3):967-973. doi:10.1161/01.STR.0000258112.14918.24.
27. Berkhemer OA, Fransen PSS, Beumer D, et al. A Randomized Trial of Intraarterial Treatment for Acute Ischemic Stroke. N Engl J Med. 2015;372(1):11-20. doi:10.1056/NEJMoa1411587.
28. Saver JL, Goyal M, Bonafe A, et al. Stent-Retriever Thrombectomy after Intravenous thrombolysis vs. thrombolysis Alone in Stroke. N Engl J Med. 2015;372(24):2285-2295. doi:10.1056/NEJMoa1415061.
29. Jovin TG, Chamorro A, Cobo E, et al. Thrombectomy within 8 Hours after Symptom Onset in Ischemic Stroke. N Engl J Med. 2015;372(24):2296-2306. doi:10.1056/NEJMoa1503780.
30. Goyal M, Menon BK, van Zwam WH, et al. Endovascular thrombectomy after large-vessel ischaemic stroke: a meta-analysis of individual patient data from five randomised trials. Lancet. 2016;387(10029):1723-1731. doi:10.1016/S0140-6736(16)00163-X.
31. Jovin TG, Saver JL, Ribo M, Pereira V, Furlan A, Bonafe A, Baxter B, Gupta R, Lopes D, Jansen O, Smith W, Gress D, Hetts S, Lewis RJ, Shields R, Berry SM, Graves TL, Malisch T, Rai A, Sheth KN, Liebeskind DS NR. Diffusion-weighted imaging or computerized

tomography perfusion assessment with clinical mismatch in the triage of wake up and late presenting strokes undergoing neurointervention with Trevo (DAWN) trial methods. Int J Stroke. 2017;12(6):641-652. doi:10.1177/1747493017710341.

Appendix A. Abbreviations

Abbreviation	Full Term
ACA	Anterior Cerebral Artery
AE	Adverse Event
AHA	American Heart Association
AIS	Acute Ischemic Stroke
AOL	Arterial Occlusive Lesion
AR	Adverse Reaction
ARU	Adverse Reaction Unexpected
ASA	American Stroke Association
ASPECT	Alberta Stroke Program Early Computed Tomography Score
AT	As Treated
BP	Blood Pressure
CA	Competent Authority
CEC	Clinical Events Committee
CIP	Clinical Investigational Plan
CRF	Case Report Form
CTA	Computerized Tomography Angiography
CTP	Computerized Tomography Perfusion
DBP	Diastolic Blood Pressure
DNR	Do Not Resuscitate
DWI	Diffusion Weighted Imaging
EC	Ethics Committee
eCRF	Electronic Case Report Form
EDC	Electronic Data Capture
ENT	Embolization to New Territory
ESO	European Stroke Organization
EV	Endovascular
EVT	Endovascular Therapy
GCP	Good Clinical Practice

IA	Intra-Arterial
ICA	Internal Carotid Artery
ICA-T	Internal Carotid Artery Terminus
ICF	Informed Consent Form
ICH	International Conference on Harmonization
ICH	Intracranial Hemorrhage
IF	Investigator's File
IFU	Instructions For Use
INR	International Normalized Ratio
IRB	Institutional Review Board
ITT	Intent-To-Treat
IV	Intravenous
IVH	Intraventricular Hemorrhage
IVRS/IWRS	Interactive Voice Response System / Interactive Web Response System
LAR	Legally Authorized Representative
LTFU	Lost To Follow Up
LVO	Large Vessel Occlusion
M1	the initial horizontal segment of the MCA, prior to the first bifurcation or trifurcation
M2	the portions of the MCA distal to the first bifurcation or trifurcation, but prior to the second bifurcation
MCA	Middle Cerebral Artery
MR/MRI	Magnetic Resonance / Magnetic Resonance Imaging
MRA	Magnetic Resonance Angiography
mRS	Modified Rankin Scale
MT	Mechanical Treatment
mTICI	Modified Thrombolysis in Cerebral Infarction
NCCT	Non Contrast Computerized Tomography
ND	Neurological Deterioration
NIHSS	National Institute of Health Stroke Scale
PP	Per Protocol

PT	Prothrombin Time
PTT	Partial Thromboplastin Time
PWI	Perfusion Weighted Imaging
RIH	Remote Intracerebral Hemorrhage
SAE	Serious Adverse Event
SAH	Subarachnoid Hemorrhage
SAR	Serious Adverse Reaction
SBP	Systolic Blood Pressure
SICH	Symptomatic Intracranial Hemorrhage
SUSAR	Suspected Unexpected Serious Adverse Reaction
TICI	Thrombolysis In Cerebral Infarction
TIMI	Thrombolysis in Myocardial Infarction
TLKW	Time Last Known Well
TMF	Trial Master File
tPA	Tissue Plasminogen Activator (alteplase)

Appendix B Definitions

Access Site Complication: Complication related to the vascular access site for the index procedure including but not limited to bleeding, hematoma, pseudo-aneurysm, tears, pain or occlusion. Some of these complications may require additional treatment such as blood transfusion or surgical repair.

Device Malfunction/Nonconformity: The failure of a device to meet its performance specifications or otherwise perform as intended. Performance specifications include all claims made in the labeling for the device. The intended performance of a device refers to the intended use for which the device is labeled or marketed.

Distal Embolization (DE): Any downstream occlusion distal to the target artery lesion (TAL), into the target ischemic territory, is considered DE unless complete angiogram or pre procedure non-invasive imaging demonstrated patency of these distal branches.

Embolization to New Territory (ENT): Embolization into a previously uninvolved area of the brain.

Epidural hemorrhage: Blood between the dura mater and the arachnoid mater.

Intracranial Hemorrhage: A hemorrhage, or bleeding, within the skull

Intracranial Hemorrhage Types – according to HEIDELBERG bleeding classification:

HI-1	Small petechiae within ischemic field without mass effect
HI-2	Confluent petechiae within ischemic field without mass effect
PH-1	Hematoma within ischemic field with some mild space-occupying effect but involving ≤ 30% of the infarcted area
PH-2	Hematoma within ischemic field with space-occupying effect involving > 30% of the infarcted area
RIH	Any intraparenchymal hemorrhage remote from the ischemic field
IVH	Intraventricular hemorrhage
Subdural	Blood between the dura mater and the arachnoid mater
Epidural	Blood between the dura mater and the skull
SAH	Subarachnoid hemorrhage

Intramural arterial dissection: A tear or damage to the inner arterial wall that occurs during the index procedure. The intramural arterial dissection may be identified angiographically as minor radiolucent area to luminal filling defect on imaging.

Intra-procedural Mortality: Death occurring during the index thrombectomy procedure

Intra-ventricular Hemorrhage (IVH): Bleeding into the brain's ventricular system.

MCA-M1 segment: is defined as the first branch of the intracranial ICA which courses horizontally from its branching point off the ICA through the sylvian fissure up to the first bifurcation distal to the lenticulo-striate arteries, in the Sylvian fissure. The absence of visualization of the origin of one of the two M2 branches will be considered as a distal M1 occlusion and therefore eligible.

Medical Management: The specific implementation of best medical management should be consistently applied and in accord with the relevant AHA or ESO guidelines, as applicable in the country of treatment.

Modified Rankin Scale: Scale for measuring the degree of disability or dependence in the daily activities of people who have suffered a stroke or other causes of neurological disability.

Neurological Deterioration: worsening in the NIHSS of ≥ 4 points compared to baseline or an NIHSS of ≥6

NIHSS Score: An assessment to objectively quantify the impairment caused by a stroke. It is composed of 11 items, each of which scores a specific ability between a 0 and 4. For each item, a score of 0 typically indicates normal function in that specific ability, while a higher score is indicative of some level of impairment. The individual scores from each item are summed in order to calculate a total NIHSS score. The maximum possible score is 42, with the minimum score being a 0.

Pre-stroke disability: Obtained at baseline, but representative of the subject's status before the index stroke, assessed by mRS on medical history obtained from subject's medical chart, or family members.

Protocol Deviation: A protocol deviation/violation is generally an unplanned excursion or non-compliance from the protocol, consent form and any other information relating to the research study that is not implemented or intended as a systematic change
Major deviation: may lead to exclusion of patients from eligibility analysis and/or their discontinuation from the study. A major deviation may impact participant safety, or its willingness to participate in the study (i.e.: enrollment without obtaining appropriate informed consent; no fulfillment of eligibility criteria; randomization irregularities; non reporting of SAE/UADE...)
Minor deviation: a deviation that does not impact subject safety, compromise the integrity of study data and/or affect subject's willingness to participate in the study

Stroke-related Death: Death related to the index stroke; to systemic complications associated with the index stroke, or a new stroke.

Subarachnoid Space: the area between the arachnoid membrane and the pia mater surrounding the brain.

Symptomatic ICH (SICH): the HEIDELBERG classification defines it as any apparently extravascular blood in the brain or within the cranium that is associated with clinical deterioration as defined by an increase of 4 points or more in the NIHSS, or that led to death and was judged to be the predominant cause of a neurologic deterioration.

Un-witnessed Strokes: Subjects with last known well time and symptoms first observed time known to be different, but not known to have symptoms first detected on awakening from sleep.

Vessel Perforation: A hole or puncture (perforation) in the vessel wall that occurs unintentionally during the index procedure. The perforations may be seen angiographically during the index procedure by frank or free extravasation of the contrast into the surrounding tissue or blush or localized contrast extending outside the vessel lumen.

Wake-up Strokes: Subjects known to have symptoms first detected on awakening from sleep.

Weighted mRS: A numerical value representing the clinical utility of each mRS category.

Witnessed Strokes: Subjects with last known well time and symptoms first observed time known to be the same.

Appendix C. Angiographic Core Lab Scales

Alberta stroke program early CT score/scale (ASPECTS)

A 10-point quantitative CT scan score used in MCA ischemic strokes. Segmental assessment of the MCA vascular territory is made and 1 point is deducted from the initial score of 10 for every region involved:

- caudate
- putamen
- internal capsule
- insular cortex
- M1: "anterior MCA cortex," corresponding to frontal operculum
- M2: "MCA cortex lateral to insular ribbon" corresponding to anterior temporal lobe
- M3: "posterior MCA cortex" corresponding to posterior temporal lobe
- M4: "anterior MCA territory immediately superior to M1"
- M5: "lateral MCA territory immediately superior to M2"
- M6: "posterior MCA territory immediately superior to M3"

TIMI Grade Scale

0 - No perfusion
1 - Penetration with minimal perfusion
2a - Partial perfusion of the artery & its main branches < 50%
2b - Partial perfusion of the artery & its main branches ≥50%
3 - Complete perfusion

Collateral Flow Grade

0 - No collaterals visible to the ischemic site
1 - Slow collaterals to the periphery of the ischemic site with persistence of some of the defect
2 - Rapid collaterals to the periphery of ischemic site with persistence of some of the defect and to only a portion of the ischemic territory
3 - Collaterals with slow but complete angiographic blood flow of the ischemic bed by the late venous phase
4 - Complete and rapid collateral blood flow to the vascular bed in the entire ischemic territory by retrograde perfusion

AOL Grade

0 - No recanalization of the primary occlusive lesion
I - Incomplete or partial recanalization of the primary occlusive lesion with no distal flow

II - Incomplete or partial recanalization of the primary occlusive lesion with any distal flow
III -
Complete recanalization of the primary occlusion with any distal flow

Modified TICI Scale (mTICI Scale)

0 - No Perfusion: No antegrade flow beyond the point of occlusion
1 - Penetration with Minimal Perfusion: The contrast material passes beyond the area of obstruction but fails to opacify the entire cerebral bed distal to the obstruction for the duration of the angiographic run
2 - Partial Perfusion: The contrast material passes beyond the obstruction and opacifies the arterial bed distal to the obstruction; However, the rate of entry of contrast into the vessel distal to the obstruction and/or its rate of clearance from the distal bed are perceptibly slower than its entry into and/or clearance from comparable areas not perfused by the previously occluded vessel, e.g., the opposite cerebral artery or the arterial bed proximal to the obstruction
2a - Only partial filling (< 50%) of the entire vascular territory is visualized
2b - Filling of ≥ 50% all of the expected vascular territory is visualized, but the filling is slower than normal
3 - Complete Perfusion: Antegrade flow into the bed distal to the obstruction occurs as promptly as into the obstruction and clearance of contrast material from the involved bed is as rapid as from an uninvolved other bed of the same vessel or the opposite cerebral artery.

Appendix D. Medico-economic study

COST-EFFECTIVENESS STUDY - IN EXTREMIS
SYNOPSIS
MOSTE MinOr Stroke Therapy Evaluation
_&_
LASTE Large Stroke Therapy Evaluation.

BACKGROUND	In France, Stroke represents the leading cause of acquired handicap in adults, (1). In 2015, it has been estimated that 500,000 people suffered from stroke in France. The annual incidence varies between 100,000 and 145,000 patients, of whom 15-20% die at the end of the first month, 50% after 5 years in case of cerebral infarctions, and 75% survive with disability [1], [2]. In addition, the socio-economic impact of stroke represents an important challenge for the global health system. Indeed, the annual cost of stroke care has been estimated at €8.9 billion in 2007 [2], accounting for 3% of health expenditures in France. The mean cost per event is estimated at €16,686 per patient in the first year, 60% of which are related to hospital care [3]. Rehabilitation and long-term care costs have rarely been investigated in the literature, but would reach €2.4 billion per year [3]. Productivity losses would account for €255.9 million per year and the income loss of patients with stroke, partially offset by the income from social insurance, is estimated at €63.3 million [3]. These findings have stressed the need to make efficient choices regarding stroke prevention and management strategies. Intravenous thrombolysis (IVT) delivered within four hours and half of stroke onset has long been considered the gold standard for stroke treatment. Nowadays, a new revascularization technique referred to as mechanical thrombectomy (MT), in combination with IVT, has been shown to be more effective [4]–[7], especially in case of stroke due to large vessel occlusion. In 2015, the French Neuroradiology Society and in 2018 the American Stroke Association and the European Stroke Organization, have recommended to perform MT in the acute phase of stroke treatment up to 6 hours after symptom onset in patients with proximal cerebral artery occlusion (carotid, middle cerebral) (grade A, level 1a) [8]. An estimated 6,000 -7,000 MT procedures are performed annually in France, 50% of them being performed in Ile-de-France [8]. In this context, the French national health authority (HAS) has made recommendations on the management of acute ischemic stroke (AIS), and has initiated a pricing process for MT techniques, stressing the need to evaluate the cost of this new technology to make efficient choice of treatment. Recently, after DAWN and DIFUSE 3 Studies, it has been proved that MT is effective after 6 hours and up to 24hours, in case of favorable mismatch profiles. IN Extremis is aiming at extending the level of evidence for MT out of the edge of the established indication, regarding the LOW NIHSS presentations 0-5 and LOW ASPECT score from 0-5. Numerous economic studies have compared Stent Retriever (SR) to IVT [9]–[13] but few studies have investigated the cost-effectiveness of MT in the last extension of the indication.

	The IN EXTREMIS study will be conducted in a context of MT pricing in France, USA and Spain. Therefore, the aim of the study will be to evaluate the cost-effectiveness of MT in patients' subgroup NIHSS 0-5 and ASPECT 0-5 compared to Standalone Medical Therapy in AIS due to large vessel occlusion.
MATERIALS AND METHODS	**1- Study design and patient characteristics** The IN EXTREMIS Trial [16] is a multicentric, prospective, randomized, controlled, open-label, blinded endpoint (PROBE) clinical trial investigating the best first-line MT in Minor Stroke NIHSS 0-5 and Large and Very Large Stroke ASPECT 0-5. This study included adult patients admitted for suspected IS secondary to occlusion of the anterior circulation, within 6 hours after last seen well in LASTE and within 24hours for MOSTE with a minimum follow for up to 90 days after the stroke onset. Economic data are collected at the same time as clinical efficacy and safety data. Thus, Electronic Case Report Form (e-CRF) will be used to collect data on medical resources (materials, devices, drugs, equipment), mTICI at the end of the procedure, and on 90-day modified Rankin score (mRS) from the time of randomization. The study is registered with ClinicalTrials.gov (NCTASOPHIE A COMPLETER) and is conducted in accordance with the tenets of the declaration of Helsinki and Good Clinical Practices. **2- Analytic overview** A cost-effectiveness analysis will be performed from the hospital perspective in the context of pricing of MT techniques in France. A micro-costing approach will be used to collect detailed medical and non-medical resources used. Procedural cost of MT will be estimated. Prices of devices and drugs – in euro (€)for the period of the trial (2019-2021) – will be collected from the pharmacy departments of participating French hospitals. Unit prices of equipment used in the operating room will be collected from the biomedical department of participating hospitals. Medical and non-medical staff times will be determined using the annual gross salaries reported in the hospital index grid, in France, Spain and US. Provisional unit costs are detailed in Table 1. Clinical efficacy data used in this analysis will be sourced from the IN EXTREMIS study. Primary outcome will be the "incremental cost/patient functionally dependent" ratio (mRS ≤2). Secondary outcome of the cost-effectiveness analysis will be the "incremental cost/patient successfully revascularized" ratio (mTICI=2b or 3). Procedure cost included cost of failure. **3- Resources utilization and costs** **a. Materials and drugs** All medical materials, devices and drugs used for each patient during surgical procedures will be recorded in the CRF. Materials will include access catheters and wires, support catheters, micro catheters and wires, and thrombectomy devices, whether it was direct MT or in case of secondary thrombectomy in MOSTE. All types of Contact Aspiration catheters FDA/CE approved and all FDA/CE approved SR were considered. Adjuvant treatments administered to patients in the operating room will be recorded. They will include

Abciximab (ReoPro, Lilly, France), Heparin (Sanofi-Aventis, France), Alteplase (Boehringer Ingelheim, France), extracranial and intracranial stenting.
The cost of materials and drugs will be calculated by multiplying the number of used resources recorded in the e-CRF by their unit price based on the information provided by the biomedical and pharmaceutical departments of each participating hospital. Prices without discount including VAT will be used. When a price will not be available from a manufacturer, the price of the same material from another manufacturer will be applied. Most unit prices will be obtained from the pharmacy department of the Montpellier Hospital (France. An operating room preparation package will be also added to this cost. This package will be calculated by the pharmacy department of the hospital and was estimated based on a micro-costing process.

b. Staff

Based on expert advices, 4 types of staff involved in surgery were identified: neuroradiologists, anesthetists, state-certified nurse anesthetists (IADE) and radiographers. They will be valued based on the time spent for surgery, i.e. the duration of the surgical procedure. A gross hourly cost including charges was calculated by dividing the annual salary by the total number of working hours per year estimated at 1,856 hours. The annual salary will be taken from the 2019 salary grid of the participating hospitals.

c. Operating room

The cost of the operating room will be taken into account in order to valorize the time the equipment was used during the surgical procedure. The operation time of each equipment will be corresponded to the duration of each intervention. Identified equipment, validated with experts, included: one angiography machine, 4 computers, 2 consoles and one injection machine. The cost of the operating room will be then calculated by multiplying the duration of each intervention by the hourly gross cost of the operating room. The hourly cost of the operating room was estimated by summing the hourly cost of all equipment mentioned above. The hourly cost of each equipment was calculated using their unit price, the depreciation rate (based on the information provided by the biomedical department of each hospital) and the annual maintenance cost. When the maintenance cost was not available, we considered that it was 20% of the equipment price according to European recommendations. Depreciation periods of 3 years for computers and injection machine and 5 years for the angiography machine were applied. We considered that all equipment of the operating room operated 19h a day. Finally, a running cost (including electricity and facility cost) corresponding to 10% of the total cost of the operating room was taken into account.

4- Efficacy outcomes

The primary outcome of the cost-effectiveness analysis will be the number of patients who will be functionally independent at 3 months (mRS≤2), based on a score ranging between 0 and 6 on the modified Rankin Scale (ranging from 0= no symptom to 6= death).

	Data on clinical efficacy will be taken from the IN EXTREMIS Trial study, two analysis will be performed on the MOSTE and the LASTE trial with a comparison between the medical group and the direct MT group. **Statistical analysis** For each treatment group, a cost per procedure will be estimated. Procedural cost per patient successfully revascularized and procedural cost per patient functionally independent will also be estimated and compared between both groups. Material and drug costs, personal cost and operating room cost will be estimated for each treatment group. Categorical variables will be reported as frequencies and percentages. Continuous variables will be reported as means (Standard Deviation) or median (Interquartile Range). Categorical variables will be compared using chi-square test. Costs will be compared between the two groups of treatment using a Wilcoxon rank-sum test. A p value <0.05 will be considered statistically significant. Statistical analyzes will be performed using SAS 9.4 software. **5- Cost-effectiveness analysis** The cost-effectiveness of the two treatment groups, for MOSTE and LASTE will be estimated using the Incremental Cost Effectiveness Ratio (ICER). The ICER will be calculated as a difference between the total treatment costs divided by the difference in the outcome. Two ICERs will be estimated according to the primary outcome (mRS ≤2). **6- Deterministic sensitivity analysis** A deterministic sensitivity analysis will be performed to investigate the impact of the increase in procedure price on the ICER.
BIBLIOGRAPHY	[1] Haute Autorité de Santé, “Prise en charge précoce (alerte, phase préhospitalière, phase hospitalière initiale, indications de la thrombolyse),” Recommandations de bonnes pratiques, Mai 2009. [2] E. Fery-Lemonnier, “La prévention et la prise en charge des accidents vasculaires cérébraux en France,” Juin 2009. [3] K. Chevreul, I. Durand-Zaleski, A. Gouépo, E. Fery-Lemonnier, M. Hommel, and F. Woimant, “Cost of stroke in France,” Eur. J. Neurol., vol. 20, no. 7, pp. 1094–1100, Jul. 2013. [4] O. A. Berkhemer et al., “A Randomized Trial of Intraarterial Treatment for Acute Ischemic Stroke,” N. Engl. J. Med., vol. 372, no. 1, pp. 11–20, Jan. 2015. [5] J. L. Saver et al., “Solitaire™ with the Intention for Thrombectomy as Primary Endovascular Treatment for Acute Ischemic Stroke (SWIFT PRIME) Trial: Protocol for a Randomized, Controlled, Multicenter Study Comparing the Solitaire Revascularization Device with IV tPA with IV tPA Alone in Acute Ischemic Stroke,” Int. J. Stroke, vol. 10, no. 3, pp. 439–448, Apr. 2015. [6] M. Goyal et al., “Randomized Assessment of Rapid Endovascular Treatment of Ischemic Stroke,” N. Engl. J. Med., vol. 372, no. 11, pp. 1019–1030, Mar. 2015. [7] T. G. Jovin et al., “Thrombectomy within 8 Hours after Symptom Onset in Ischemic Stroke,” N. Engl. J. Med., vol. 372, no. 24, pp. 2296–2306, Jun. 2015. [8] Haute Autorité de la Santé, “Thrombectomie des artères intracrâniennes par voie endovasculaires,” France, Rapport d’évaluation technologique, Nov. 2016. [9] T. I. Shireman et al., “Cost-Effectiveness of Solitaire Stent Retriever Thrombectomy for Acute Ischemic Stroke: Results From the SWIFT-PRIME Trial (Solitaire With the Intention for

Thrombectomy as Primary Endovascular Treatment for Acute Ischemic Stroke),” Stroke, vol. 48, no. 2, pp. 379–387, Feb. 2017.
[10] J. Ganesalingam, E. Pizzo, S. Morris, T. Sunderland, D. Ames, and K. Lobotesis, “Cost-Utility Analysis of Mechanical Thrombectomy Using Stent Retrievers in Acute Ischemic Stroke,” Stroke, vol. 46, no. 9, pp. 2591–2598, Sep. 2015.
[11] X. Xie et al., “Mechanical thrombectomy in patients with acute ischemic stroke: a cost-utility analysis,” CMAJ Open, vol. 4, no. 2, pp. E316–E325, Jun. 2016.
[12] K. Lobotesis, R. Veltkamp, I. H. Carpenter, L. M. Claxton, J. L. Saver, and R. Hodgson, “Cost-effectiveness of stent-retriever thrombectomy in combination with IV t-PA compared with IV t-PA alone for acute ischemic stroke in the UK,” J. Med. Econ., vol. 19, no. 8, pp. 785–794, Aug. 2016.
[13] A. S. Kim, M. Nguyen-Huynh, and S. C. Johnston, “A Cost-Utility Analysis of Mechanical Thrombectomy as an Adjunct to Intravenous Tissue-Type Plasminogen Activator for Acute Large-Vessel Ischemic Stroke,” Stroke, vol. 42, no. 7, pp. 2013–2018, Jul. 2011.
[14] A. Comai, T. Haglmüller, F. Ferro, E. Dall’Ora, R. Currò Dossi, and G. Bonatti, “Sequential endovascular thrombectomy approach (SETA) to acute ischemic stroke: preliminary single-centre results and cost analysis,” Radiol. Med. (Torino), vol. 120, no. 7, pp. 655–661, Jul. 2015.
[15] A. S. Turk et al., “ADAPT FAST study: a direct aspiration first pass technique for acute stroke thrombectomy,” J. NeuroInterventional Surg., vol. 6, no. 4, pp. 260–264, May 2014.
[16] B. Lapergue et al., “Effect of Endovascular Contact Aspiration vs Stent Retriever on Revascularization in Patients With Acute Ischemic Stroke and Large Vessel Occlusion: The ASTER Randomized Clinical Trial,” JAMA, vol. 318, no. 5, p. 443, Aug. 2017.

APPENDICES

Table1: Exemple of Units cost of equipments, consumables, devices and personal(€,)

	Unit price (€, TTC)[a]	Sources
Medical acces devices		
Diagnostic catheter	€23	Pharmacy
Support catheter1	€44	Pharmacy
Support catheter2	€77	Pharmacy
Micro catheters	€369-€850	Pharmacy
Micro support catheter	€390	Pharmacy
Long catheters	€220-€630	Pharmacy
Thrombectomy devices		
Stent Retriever	€3 840-€4 958	Pharmacy and Manufacturer
Contact Aspiration catheter	€3 000	Pharmacy and Manufacturer
Operating room equipment		
Computer	€1 543	Biomedical unit
Angiography machine	€1 374 000	Biomedical unit
Computer monitor	€5 680	IT unit
Machine of contrast product	€29 578	Biomedical unit
Adjuvant treatment		
Reopro	€360	Pharmacy
actylise	€158	Pharmacy
Heparin	€2	Pharmacy
Extracranial stenting	€967	Pharmacy

Intracranial stenting	€4 500	Pharmacy
Medical and non medical staff		
Neurointerventionists	€70	Salary grid of public hospital service in France
Anesthetists	€67	Salary grid of public hospital service in France
Nurse anesthetists	€29	Salary grid of public hospital service in France
Radiographers	€27	Salary grid of public hospital service in France

[a]Prices are with tax and are not discounted

Appendix E. Adverse Events Management in Spain

Adverse Event Definitions and Classification

Adverse event
Safety reporting in clinical investigations of medical devices under the Regulation (EU) 2017/745
Any untoward medical occurrence, unintended disease or injury or any untoward clinical signs, including an abnormal laboratory finding, in subjects, users or other persons, in the context of a clinical investigation, whether or not related to the investigational device.

Adverse device effect (ADE)
Any adverse event related to the use of an investigational medical device or a comparator.

Serious Adverse event or reaction:
Any adverse event that led to any of the following:
a) death,
b) serious deterioration in the health of the subject, that resulted in any of the following:
i. life-threatening illness or injury,
ii. permanent impairment of a body structure or a body function,
iii. hospitalisation or prolongation of patient hospitalisation,
iv. medical or surgical intervention to prevent life-threatening illness or injury or permanent impairment to a body structure or a body function,
v. chronic disease,
c) foetal distress, foetal death or a congenital physical or mental impairment or birth defect

Serious adverse device effect (SADE)
Any adverse device effect that has resulted in any of the consequences characteristic of a serious adverse event.

New Finding
New information discovered as the result of an inquiry/investigation/test based on the occurrence of the event. Follow-up from the event.
'Device deficiency'
Any inadequacy in the identity, quality, durability, reliability, safety or performance of an investigational device, including malfunction, use errors or inadequacy in information supplied by the manufacturer.

Anticipated serious adverse device effect (ASADE)
Any serious adverse device effect which by its nature, incidence, severity or outcome has been identified in the last risk assessment document upon serious adverse device effect occurred.

Causality

Not related: Relationship to the device, comparator or procedures can be excluded when:
- the event has no temporal relationship with the use of the investigational device, or the procedures related to application of the investigational device;
- the serious adverse event does not follow a known response pattern to the medical device (if the response pattern is previously known) and is biologically implausible;
- the discontinuation of medical device application or the reduction of the level of activation/exposure - when clinically feasible - and reintroduction of its use (or increase of the level of activation/exposure), do not impact on the serious adverse event;
- the event involves a body-site or an organ that cannot be affected by the device or procedure;
- the serious adverse event can be attributed to another cause (e.g. an underlying or concurrent illness/ clinical condition, an effect of another device, drug, treatment or other risk factors);
- the event does not depend on a false result given by the investigational device used for diagnosis9, when applicable;

In order to establish the non-relatedness, not all the criteria listed above might be met at the same time, depending on the type of device/procedures and the serious adverse event.

Possible: The relationship with the use of the investigational device or comparator, or the relationship with procedures, is weak but cannot be ruled out completely. Alternative causes are also possible (e.g. an underlying or concurrent illness/ clinical condition or/and an effect of another device, drug or treatment). Cases where relatedness cannot be assessed, or no information has been obtained should also be classified as possible.

Probable: The relationship with the use of the investigational device or comparator, or the relationship with procedures, seems relevant and/or the event cannot be reasonably explained by another cause.

Causal relationship: the serious adverse event is associated with the investigational device, comparator or with procedures beyond reasonable doubt when:
- the event is a known side effect of the product category the device belongs to or of similar devices and procedures;
- the event has a temporal relationship with investigational device use/application or procedures;
- the event involves a body-site or organ that
    - o the investigational device or procedures are applied to;
    - o the investigational device or procedures have an effect on;
- the serious adverse event follows a known response pattern to the medical device (if the response pattern is previously known);
- the discontinuation of medical device application (or reduction of the level of activation/exposure) and reintroduction of its use (or increase of the level of activation/exposure), impact on the serious adverse event (when clinically feasible);

- other possible causes (e.g. an underlying or concurrent illness/ clinical condition or/and an effect of another device, drug or treatment) have been adequately ruled out;

- harm to the subject is due to error in use;

- the event depends on a false result given by the investigational device used for diagnosis10, when applicable;

In order to establish the relatedness, not all the criteria listed above might be met at the same time, depending on the type of device/procedures and the serious adverse event.
The sponsor and the investigators will distinguish between the serious adverse events related to the investigational device and those related to the procedures (any procedure specific to the clinical investigation). An adverse event can be related both to procedures and the investigational device. Complications caused by concomitant treatments not imposed by the clinical investigation plan are considered not related. Similarly, several routine diagnostic or patient management procedures are applied to patients regardless of the clinical investigation plan. If routine procedures are not imposed by the clinical investigation plan, complications caused by them are also considered not related.
The relationship between a serious adverse event and the procedure or the device, needs to be assessed separately. This does however not mean that they are mutually exclusive, a serious adverse event can be related to both the procedure and the device, or it can be related only to the procedure or only to the device. When it is unclear whether an event is related to the device or to the procedure, the investigator should:
• set the Relationship to device to possible (or higher)
AND
• set the Relationship to procedure to possible (or higher)

Since it is the healthcare provider who performs the procedures and manages/handles the medical device(s), the causality assessment of this healthcare provider should prevail.
In some particular cases the event may not be adequately assessed because information is insufficient or contradictory and/or the data cannot be verified or supplemented. The sponsor and the Investigators will make the maximum effort to define and categorize the event and avoid these situations. Where an investigator assessment is not available and/or the sponsor remains uncertain about classifying the serious adverse event, the sponsor should not exclude the relatedness; the event should be classified as "possible" and the reporting not be delayed.
Particular attention shall be given to the causality evaluation of unanticipated serious adverse events. The occurrence of unanticipated events related could suggest that the clinical investigation places subjects at increased risk of harm than was to be expected beforehand.

Events of interest
See section 20.3 in the protocol

Investigator responsibilities

Collection of AEs

Complete and appropriate data on all AEs and device deficiency (DD) (which could have resulted in a serious adverse event) experienced during the clinical trial will be recorded on the AE form and the DD form of the CRF on an ongoing basis for the duration of the study. Each AE/DD report shall include a description of the event/DD, an assessment of its seriousness according to the criteria listed above, its duration, intensity, relationship to the study devices, other causality factors (if any), any concomitant medication dispensed, actions taken with the study drug or other therapeutic interventions and outcome at the end of the observation period.
For each AE/DD, a separate AE/DD form will be filled in.
AEs/DD will be followed up until their resolution or stabilisation. However, the observation period will be cut off after the last patient has finished his final visit (LPLV)
All AEs/DD and the treatment and follow-up required must be documented in the subject's medical records and in the eCRF.

Exception to collection
The following circumstances will not be collected:
- ✓ Admission for social or administrative reason
- ✓ Hospitalization predefined by the protocol
- ✓ Hospitalization for medical or surgical treatment planned before the research

Notification of SAEs/DD
The investigator evaluates each adverse event in terms of severity and each DD which could have resulted in a SAE.
The investigator must notify to the Sponsor all serious adverse events or all DD which could have resulted in a SAE, **without delay**, the day the event is known, if it occurs:
- ✓ At the **time of enrollment** (defined as **time of randomization**)
- ✓ During all the duration of the patient participation

Exception to notification
Some serious adverse events are not requiring immediate reporting but only be documented on the CRF/ source data as AE.
Underlying (pre-existing) symptoms or diseases are not reported as Adverse Events (AEs) unless there is an increase in severity or frequency during the course of the investigation, but they need to be reported in the eCRF as Relevant Medical History

Summary table of the notification circuit by type of event

Sort of event	**Notification procedures**	**Period of the notification to the Sponsor**
Non serious adverse event	In the CRF	No notification
Serious adverse event or event of interest DD which could have resulted in a SAE	SAE/DD notification form and follow up if necessary + Report in the CRF	**Notification without delay to the Sponsor**

New fact	Written report	**Notification without delay at the Sponsor**

In case of notification, the declaration has to be made with a serious adverse event declaration form and send by fax or email at:

Direction de la recherche et de l'Innovation
Vigilance des Essais Cliniques
Perrine ROBIN
Fax : +33 4 67 33 54 11
Vigilance-ec@chu-montpellier.fr

This initial notification is a written report and must be followed, if necessary, by detailed written supplementary report(s).

Sponsor responsibilities
Assessment of Serious Adverse Events
The assessment of the relationship between a SAE and the administration of experimental procedure is a clinical decision based on all available information at the time of, and after the occurrence of the event. This assessment has to be done by sponsor too.
In case of positive relationship between the event and the experimental procedure, the sponsor will determine whether the events are expected or unexpected based on the reference safety document.

Expedited reporting
For the purpose of this guidance and based on the definitions above, the following events are considered reportable events in accordance with MDR Art. 80(2):
a) any serious adverse event that has a causal relationship with the investigational device, the comparator or the investigation procedure or where such causal relationship is reasonably possible;

b) any device deficiency that might have led to a serious adverse event if appropriate action had not been taken, intervention had not occurred, or circumstances had been less fortunate;

c) any new findings in relation to any event referred to in points a) and b).

Serious adverse events related to a CE marked device which is part of the investigation procedure (for example a CE-marked implanting tool used in combination with a non-CE marked investigational device) are reportable per MDR Article 80(2) if there is a causal, (or reasonably possible) relationship to the device, the comparator or the investigation procedure. The reporting procedures described in this guide should then be followed in addition to the normal vigilance reporting procedures for CE marked devices.

All causality assessments should be made using the guidance in section 9. Only causality level 1 (i.e. “not related”) is excluded from reporting. If either the sponsor or the investigator has assigned a higher causality level than "not related", the event should be reported.

Annual safety report (ASR)

In addition to the expedited reporting, the sponsor shall submit, once a year, throughout the clinical trial or on request, a safety report to the competent authorities and the Ethics Committee of the concerned Member States taking into account all new available safety information received during the reporting period.
The aim of the ASR is to assess the safety conditions of subjects included in the concerned trial(s).
It should be the same for the competent authorities concerned and the Ethics Committee concerned.
The Sponsor is responsible for the ongoing safety evaluation of the Investigational Procedure.
The Sponsor should promptly notify all concerned investigators and the regulatory authorities of findings that could affect adversely the safety of subjects, impact the conduct of the trial, or alter the CA approval/ EC's favourable opinion to continue the trial.
Data Safety monitoring Committee
The DSMC ou DMC (Data Monitoring Committee) is an advisory committee that provides guidance to the sponsor with respect to the benefit - risk balance and the conduct of the clinical trial.
Its members are not involved in the study and are chosen for their competence in one or more of the investigative fields of the study in collaboration with the coordinating investigators.
The DMC is composed of at least:
· a clinician specialized in the pathology studied to evaluate the clinical aspects of tolerance and efficacy of the study
· a methodologist or statistician.

The opinion of the DMC will be sent to the sponsor in a timely manner. The sponsor will forward this opinion to the coordinating investigator, the Ethic Committee and the CA as part of the safety annual report.